



Project Progress Report

RESTAURANT MANAGEMENT WEB APPLICATION

Framework	Django (Python 3.x)
Database	MySQL Restaurant DB
Architecture	MVT (Model-View-Template)
Modules	7 functional modules
Team Size	6 students

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1. Introduction

This report documents the development of a Restaurant Management Web Application built using the Django framework (Python 3.x) and a MySQL relational database comprising 23 tables.

- The project was conceived to digitalize and centralize the core operations of a restaurant, including order management, inventory tracking, human resources, delivery coordination, and financial reporting.
- The application follows the Model-View-Template (MVT) architectural pattern native to Django, ensuring a clean separation of concerns between data models, business logic, and presentation. Seven functional modules were implemented to cover the full operational scope of a modern restaurant.

This report focuses specifically on Phases 2 through 5 of the development life cycle, covering the following key milestones:

- Phase 2 Defining all Django models that map to the 23 existing SQL tables.
- Phase 3 Implementing authentication, custom user roles, and role-based access control.
- Phase 4 Building the shared HTML template system and the base interface layout.
- Phase 5 Developing the views, URL routing, and core business logic for each module.

In addition, this report includes a Project Management section presenting the Gantt chart and class diagrams used to plan and design the system.

- A dedicated section on challenges encountered during development and the solutions applied is also provided, followed by a conclusion summarizing outcomes and lessons learned.

2. Phase 2: Django Models (23 SQL Tables in Python)

Phase 2 is the most foundational phase of the project. It involves translating all 23 existing MySQL tables into Python model classes within Django. Since the database tables were already created and populated via the Base_Restaurant.sql file, each model class was configured with `managed = False`, which instructs Django not to attempt to recreate or migrate these tables it simply maps to them.

2.1 Orders App Models

The orders app handles the core transactional data of the restaurant: clients, tables, reservations, orders, and order line items. The following models were defined in `orders/models.py`:

- Client Stores customer identity and contact details.
- TableRestaurant Maps to the SQL 'Table' entity (renamed to avoid conflict with Python's reserved word 'Table').
- Reservation Links a client to a table with date, time, and status.
- Commande Represents a full order with total amount and payment method.
- LigneCommande A junction table for order line items, linking orders to products with quantity and unit price.

2.2 Products App Models

The products app manages recipes, ingredients, and the composition of each dish. Three models were defined in `products/models.py`:

- Ingredient Represents a raw ingredient with its unit of measure and type.
- Produit A dish or product, linked to the employee who created it, with cooking time and serving count.
- CompositionProduit A junction table linking products to their ingredients with quantities used, implementing a composite primary key.

2.3 Inventory, HR, Delivery & Finance Models

The remaining four app modules each map to their respective SQL tables. The table below summarizes each app's models and key implementation notes:

App File	SQL Tables	Key Points
inventory/models.py	Stock, Variation_Stock, Approvisionnement, Fournisseur	Stock uses id_Ingredient as primary key (same as Ingredient PK)
hr/models.py	Employe, Poste, Affectation, Supervision	Affectation has a composite PK: (id_Employe, id_Poste)
delivery/models.py	Vehicule, Deplacement, Livraison	Deplacement has id_Chauffeur as FK to Employe
finance/models.py	Facture, Equipement, Maintenance, Action_Marketing	Facture.Type distinguishes client invoices from supply/salary/marketing records

For models with composite primary keys (such as Affectation), the `unique_together` Meta option was used to enforce uniqueness across the two foreign key fields, as Django does not natively support composite primary keys in the traditional sense.

```
models.py
C:\Users\hp\Desktop\PROJECT 1\Restaurant-Management-App\products\models.py
1 from django.db import models
2 # Create your models here.
3 class Ingredient(models.Model):
4     id_ingredient = models.AutoField(primary_key=True, db_column='id_Ingredient')
5     nom = models.CharField(max_length=45, db_column='Nom')
6     unite_de_mesure = models.CharField(max_length=45, db_column='Unite_de_Mesure')
7     type = models.CharField(max_length=30, db_column='Type')
8     class Meta:
9         managed = False
10        db_table = 'Ingredient'
11    def __str__(self): return self.nom
12
13 class Produit(models.Model):
14     id_produit = models.AutoField(primary_key=True, db_column='id_Produit')
15     id_createur = models.ForeignKey('hr.Employe', on_delete=models.SET_NULL, null=True, db_column='id_Createur')
16     nom = models.CharField(max_length=30, db_column='Nom')
17     description = models.CharField(max_length=45, db_column='Description')
18     duree_cuisson = models.CharField(max_length=20, db_column='Duree_Cuisson')
19     nombre_personnes = models.IntegerField(db_column='Nombre_Personnes')
20     class Meta:
21         managed = False
22         db_table = 'Produit'
23     def __str__(self): return self.nom
24
25 class CompositionProduit(models.Model):
26     id_produit = models.ForeignKey(Produit, on_delete=models.CASCADE, db_column='id_Produit')
27     id_ingredient = models.ForeignKey(Ingredient, on_delete=models.CASCADE, db_column='id_Ingredient')
28     quantite_utilisee = models.DecimalField(max_digits=10, decimal_places=2, db_column='Quantite_Utilisee')
29     class Meta:
30         managed = False
31         db_table = 'Composition_Produit'
32         unique_together = [('id_produit', 'id_ingredient')]
```



```
models.py U X
C: > Users > hp > Desktop > PROJECT 1 > Restaurant-Management-App > orders > models.py > TableRestaurant > Meta
1 from django.db import models
2
3 # Create your models here.
4 class Client(models.Model):
5     id_client = models.AutoField(primary_key=True, db_column='id_Client')
6     nom = models.CharField(max_length=20, db_column='Nom')
7     prenom = models.CharField(max_length=20, db_column='Prenom')
8     tel = models.CharField(max_length=15, db_column='Tel')
9     email = models.CharField(max_length=45, db_column='Email')
10    type_de_client = models.CharField(max_length=20, db_column='Type_de_Client')
11    class Meta:
12        managed = False
13        db_table = 'Client'
14    def __str__(self):
15        return f'{self.prenom} {self.nom}'
16
17 class TableRestaurant(models.Model): # 'Table' is a reserved word!
18    id_table = models.AutoField(primary_key=True, db_column='id_Table')
19    numero_table = models.CharField(max_length=3, db_column='Numero_Table')
20    capacite = models.IntegerField(db_column='Capacite')
21    commentaire = models.TextField(db_column='Commentaire', blank=True, null=True)
22    class Meta:
23        managed = False
24        db_table = 'Table'
25    def __str__(self):
26        return f'Table {self.numero_table}'
27
28 class Reservation(models.Model):
29    id_reservation = models.AutoField(primary_key=True, db_column='id_Reservation')
30    id_client = models.ForeignKey(Client, on_delete=models.CASCADE, db_column='id_Client')
31    id_table = models.ForeignKey(TableRestaurant, on_delete=models.CASCADE, db_column='id_Table')
32    date = models.DateField(db_column='Date')
33    heure = models.TimeField(db_column='Heure')
34    statut = models.CharField(max_length=45, db_column='Statut')
35    class Meta:
36        managed = False
37        db_table = 'Reservation'
38
39 class Commande(models.Model):
40    id_commande = models.AutoField(primary_key=True, db_column='id_Commande')
41    id_client = models.ForeignKey(Client, on_delete=models.CASCADE, db_column='id_Client')
42    date = models.DateTimeField(db_column='Date')
43    type = models.CharField(max_length=30, db_column='Type')
44    montant_total = models.IntegerField(db_column='Montant_Total')
45    mode_de_paiement = models.CharField(max_length=45, db_column='Mode_de_Paiement')
46    class Meta:
47        managed = False
48        db_table = 'Commande'
49    def __str__(self):
50        return f'Commande #{self.id_commande}'
51
52 class LigneCommande(models.Model):
53    id_commande = models.ForeignKey(Commande, on_delete=models.CASCADE, db_column='id_Commande')
54    id_produit = models.ForeignKey('products.Produit', on_delete=models.CASCADE, db_column='id_Produit')
55    quantite = models.DecimalField(max_digits=10, decimal_places=2, db_column='Quantite')
56    prix_unitaire = models.IntegerField(db_column='Prix_Unitaire')
57    class Meta:
58        managed = False
59        db_table = 'Ligne_Commande'
60        unique_together = [('id_commande', 'id_produit')]
61
62
```

3. Phase 3: Authentication & User Roles

Django provides a built-in authentication system that was extended in this phase to support six distinct user roles corresponding to the restaurant's organizational structure. Rather than using Django's default User model, a custom user model (CustomUser) was created to add role-based access control.

3.1 Custom User Model

The CustomUser class extends Django's AbstractUser and introduces a role field with six predefined choices. It also includes an optional link to the Employe table via an integer field:

Role Key	Display Name
admin	Administrateur
director	Directeur
head_chef	Chef Cuisinier
cook	Cuisinier
server	Serveur
stock_manager	Gestionnaire de Stock

Django has a built-in authentication system. You will extend it to support the 6 roles defined in the project.

Step 11: Create the Custom User Model

In the users app, create a custom user that links to an Employe and has a role field:

```
# users/models.py
from django.contrib.auth.models import AbstractUser
from django.db import models

class CustomUser(AbstractUser):
    ROLES = [
        ('admin', 'Administrateur'),
        ('director', 'Directeur'),
        ('head_chef', 'Chef Cuisinier'),
        ('cook', 'Cuisinier'),
        ('server', 'Serveur'),
        ('stock_manager', 'Gestionnaire de Stock'),
    ]
    role = models.CharField(max_length=20, choices=ROLES, default='server')
    # Optional: link to your Employe table
    employe_id = models.IntegerField(null=True, blank=True)

    def __str__(self):
        return f'{self.username} ({self.get_role_display()})'
```

Then in settings.py, add this line so Django uses your custom user:

```
AUTH_USER_MODEL = 'users.CustomUser'
```

Step 12: Run the First Migration

This creates the users table (and Django's internal tables) in your MySQL database:

```
python manage.py makemigrations users
python manage.py migrate
python manage.py createsuperuser # Create your admin account
```

Step 13: Create Login and Logout Views

Django already has built-in login/logout views. In users/urls.py:

```
# users/urls.py
from django.contrib.auth import views as auth_views
from django.urls import path

urlpatterns = [
    path('login/', auth_views.LoginView.as_view(template_name='users/login.html'), name='login'),
    path('logout/', auth_views.LogoutView.as_view(), name='logout'),
]
```

Step 14: Create Role-Based Permission Decorators

Create a file `users/decorators.py` that you will use to protect any view by role:

```
# users/decorators.py
from django.contrib.auth.decorators import login_required
from django.core.exceptions import PermissionDenied
from functools import wraps

def role_required(*roles):
    def decorator(view_func):
        @wraps(view_func)
        @login_required
        def _wrapped_view(request, *args, **kwargs):
            if request.user.role in roles:
                return view_func(request, *args, **kwargs)
            raise PermissionDenied
        return _wrapped_view
    return decorator

# Usage in any view:
# @role_required('admin', 'director')
# def delete_employee(request, pk): ...
```

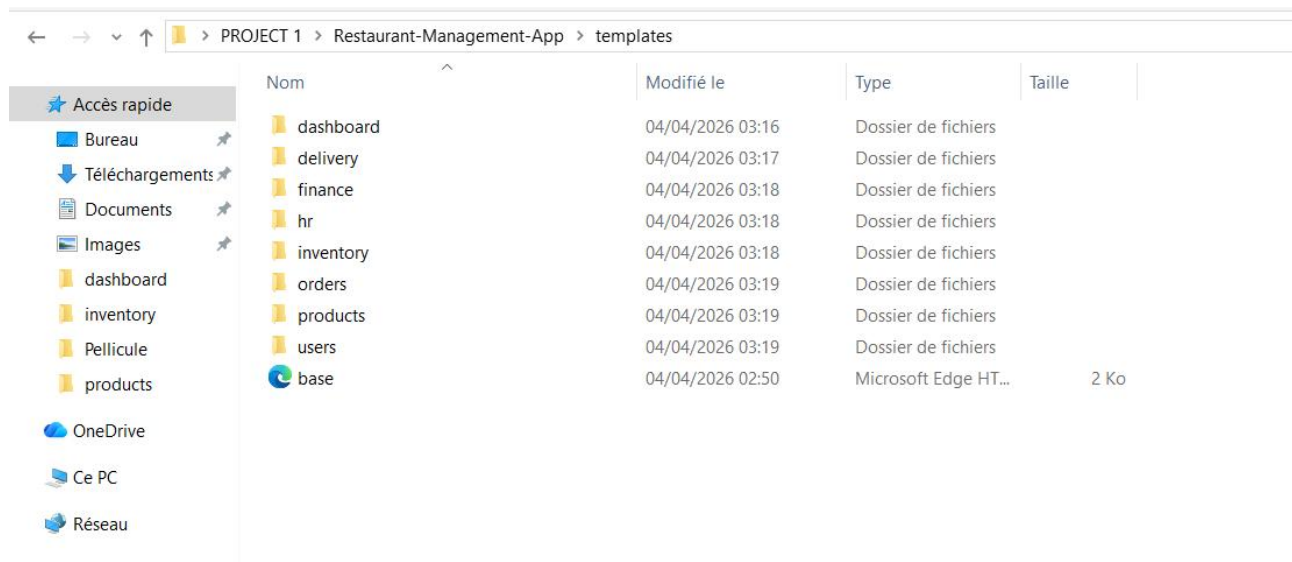
4. Phase 4 Templates & Base Interface

Before building any module-specific pages, Phase 4 establishes the shared HTML layout infrastructure that all pages will inherit.

- This follows Django's template inheritance pattern, where a single base template defines the overall page structure, and child templates override specific content blocks.

4.1 Static Files & Template Folder Structure

Two key configuration settings were added to settings.py: `TEMPLATES` (to specify the global templates directory) and `STATICFILES_DIRS` (to locate CSS and JavaScript assets). The following folder structure was created manually at the project root:



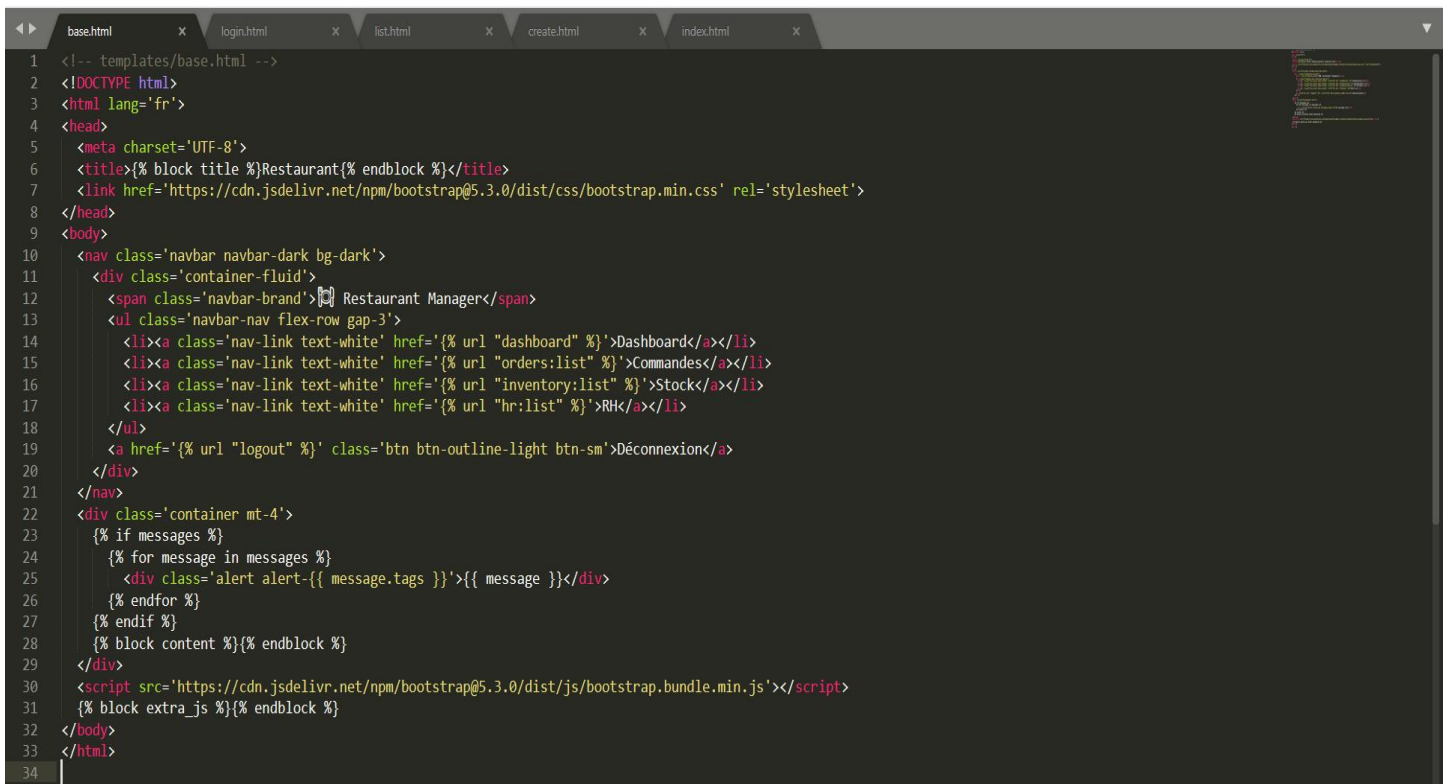
Nom	Modifié le	Type	Taille
dashboard	04/04/2026 03:16	Dossier de fichiers	
delivery	04/04/2026 03:17	Dossier de fichiers	
finance	04/04/2026 03:18	Dossier de fichiers	
hr	04/04/2026 03:18	Dossier de fichiers	
inventory	04/04/2026 03:18	Dossier de fichiers	
orders	04/04/2026 03:19	Dossier de fichiers	
products	04/04/2026 03:19	Dossier de fichiers	
users	04/04/2026 03:19	Dossier de fichiers	
base	04/04/2026 02:50	Microsoft Edge HT...	2 Ko

Each app has its own subdirectory inside templates/, separating HTML files by module. The static/ directory holds shared CSS and JS files accessible from any template using Django's `{% static %}` template tag.

4.2 Base Template (base.html)

The base.html file is the master layout for the entire application. It was built using Bootstrap 5, loaded directly from CDN to avoid any local installation. The template defines the following structural blocks :

- Navigation bar Displays the app name and links to major modules (Dashboard, Commandes, Stock, RH). The logout button is positioned at the right end.
- Flash messages block Automatically renders Django's message framework alerts (success, error, warning) between the navbar and the main content area.
- Content block An empty `{% block content %}` that each child template overrides with its specific page content.
- Extra JavaScript block An empty `{% block extra_js %}` placed at the bottom, allowing child templates to add page-specific scripts without polluting the global head.



```
1 <!-- templates/base.html -->
2 <!DOCTYPE html>
3 <html lang='fr'>
4 <head>
5   <meta charset='UTF-8'>
6   <title>{% block title %}Restaurant{% endblock %}</title>
7   <link href='https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css' rel='stylesheet'>
8 </head>
9 <body>
10   <nav class='navbar navbar-dark bg-dark'>
11     <div class='container-fluid'>
12       <span class='navbar-brand'>Restaurant Manager</span>
13       <ul class='navbar-nav flex-row gap-3'>
14         <li><a class='nav-link text-white' href='{url "dashboard" %}'>Dashboard</a></li>
15         <li><a class='nav-link text-white' href='{url "orders:list" %}'>Commandes</a></li>
16         <li><a class='nav-link text-white' href='{url "inventory:list" %}'>Stock</a></li>
17         <li><a class='nav-link text-white' href='{url "hr:list" %}'>RH</a></li>
18       </ul>
19       <a href='{url "logout" %}' class='btn btn-outline-light btn-sm'>Déconnexion</a>
20     </div>
21   </nav>
22   <div class='container mt-4'>
23     {% if messages %}
24       {% for message in messages %}
25         <div class='alert alert-{{ message.tags }}'>{{ message }}</div>
26       {% endfor %}
27     {% endif %}
28     {% block content %}{% endblock %}
29   </div>
30   <script src='https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js'></script>
31   {% block extra_js %}{% endblock %}
32 </body>
33 </html>
```

4.3 Child Template Pattern

Every other HTML page in the application starts with `{% extends 'base.html' %}` and overrides the `{% block content %}` and optionally `{% block title %}`. This ensures visual consistency across all pages while keeping each template focused only on its specific content.

As a demonstration, the orders/list.html template extends base.html and renders a Bootstrap-styled table of all orders, with columns for ID, client, date, total amount, and an action button linking to the order detail view.

```
base.html x login.html x list.html x create.html x index.html x
1 <!-- templates/orders/list.html -->
2 {% extends 'base.html' %}
3 {% block title %}Liste des Commandes{% endblock %}
4 {% block content %}
5 <h2>Commandes</h2>
6 <table class='table table-striped'>
7   <thead>
8     <tr><th>ID</th><th>Client</th><th>Date</th><th>Montant</th><th>Actions</th></tr>
9   </thead>
10  <tbody>
11    {% for commande in commandes %}
12      <tr>
13        <td>{{ commande.id_commande }}</td>
14        <td>{{ commande.id_client }}</td>
15        <td>{{ commande.date }}</td>
16        <td>{{ commande.montant_total }} FCFA</td>
17        <td><a href='{% url "orders:detail" commande.id_commande %}' class='btn btn-sm btn-info'>Voir</a></td>
18      </tr>
19    {% endfor %}
20  </tbody>
21 </table>
22 {% endblock %}
23
```

5. Phase 5 Views, URLs & Business Logic

Phase 5 is where the application becomes fully functional. Each module receives its views (the Python functions that process HTTP requests), URL patterns (the routes that map URLs to views), and business logic (the rules that govern data processing).

- Every module follows the same structural pattern: list, detail, create, edit, and delete.

5.1 Standard CRUD Pattern Orders Module

The Orders module serves as the reference implementation for the CRUD pattern applied across all modules. It involves four key views and an auxiliary helper function:

1. `order_list` Retrieves all orders sorted by date (most recent first) and renders them in the list template.
2. `order_detail` Retrieves a single order by primary key and its associated line items, then renders the detail view.
3. `order_create` Handles both GET (display the creation form) and POST (process form submission). On POST, it creates the `Commande` record, iterates over submitted line items to create `LigneCommande` records, calculates the total amount, and triggers automatic stock deduction.
4. `deduct_stock_for_product` A helper function called inside `order_create` that retrieves the product's ingredient composition and reduces each ingredient's stock quantity accordingly. It also logs a 'sortie' entry in the `VariationStock` table for traceability.


```
urls.py X
C: > Users > hp > Desktop > PROJECT 1 > Restaurant-Management-App > templates > users > urls.py > ...

1  # orders/urls.py
2  from django.urls import path
3  from . import views
4
5  app_name = 'orders'
6  urlpatterns = [
7      path('', views.order_list, name='list'),
8      path('<int:pk>/', views.order_detail, name='detail'),
9      path('new/', views.order_create, name='create'),
10 ]
11
12 # restaurant_project/urls.py (main router)
13 from django.contrib import admin
14 from django.urls import path, include
15
16 urlpatterns = [
17     path('admin/', admin.site.urls),
18     path('accounts/', include('users.urls')),
19     path('orders/', include('orders.urls')),
20     path('products/', include('products.urls')),
21     path('inventory/', include('inventory.urls')),
22     path('hr/', include('hr.urls')),
23     path('delivery/', include('delivery.urls')),
24     path('finance/', include('finance.urls')),
25     path('', include('dashboard.urls')),
26 ]
27 |
```

```
views.py 2, U X
C: > Users > hp > Desktop > PROJECT 1 > Restaurant-Management-App > templates > dashboard > views.py > ...

1  # Create your views here.
2  from django.shortcuts import render, get_object_or_404, redirect
3  from django.contrib.auth.decorators import login_required
4  from django.contrib import messages
5  from django.db import transaction
6  from .models import Commande, LigneCommande, Client
7  from inventory.models import Stock, VariationStock
8  from products.models import CompositionProduit
9  import datetime

Windsurf: Refactor | Explain | Generate Docstring | X
11 @login_required
12 def order_list(request):
13     commandes = Commande.objects.all().order_by('-date')
14     return render(request, 'orders/list.html', {'commandes': commandes})
15
Windsurf: Refactor | Explain | Generate Docstring | X
16 @login_required
17 def order_detail(request, pk):
18     commande = get_object_or_404(Commande, pk=pk)
19     lignes = LigneCommande.objects.filter(id_commande=commande)
20     return render(request, 'orders/detail.html', {'commande': commande, 'lignes': lignes})
21
Windsurf: Refactor | Explain | Generate Docstring | X
22 @login_required
23 def order_create(request):
24     if request.method == 'POST':
25         # 1. Create the order
26         commande = Commande.objects.create(
27             id_client_id=request.POST['client_id'],
28             date=datetime.datetime.now(),
29             type=request.POST['type'],
30             montant_total=0,
31             mode_de_paiement=request.POST['paiement'],
32         )
33         total = 0
34         # 2. Add each line item
35         for produit_id, qty in zip(request.POST.getlist('produit_id'), request.POST.getlist('qty')):
36             prix = int(request.POST.get(f'prix_{produit_id}', 0))
37             LigneCommande.objects.create(
38                 id_commande=commande, id_produit_id=produit_id,
39                 quantite=qty, prix_unitaire=prix
40             )
41             total += prix * float(qty)
42             # 3. Deduct ingredients from Stock automatically
43             deduct_stock_for_product(produit_id, float(qty))
44             # 4. Update total
45             commande.montant_total = int(total)
46             commande.save()
47             messages.success(request, 'Commande créée avec succès!')
48             return redirect('orders:detail', pk=commande.pk)
49     clients = Client.objects.all()
50     return render(request, 'orders/create.html', {'clients': clients})
51
Windsurf: Refactor | Explain | X
52 def deduct_stock_for_product(produit_id, qty_ordered):
53     '''Automatically reduce stock when an order is placed'''
54     compositions = CompositionProduit.objects.filter(id_produit_id=produit_id)
55     for comp in compositions:
56         stock = Stock.objects.get(id_ingredient=comp.id_ingredient)
57         qty_used = comp.quantite_utilisee * qty_ordered
58         stock.quantite_actuelle -= qty_used
59         stock.save()
60         # Log the variation
61         VariationStock.objects.create(
62             id_ingredient=comp.id_ingredient,
63             date=datetime.date.today(),
64             type='sortie',
65             quantite=qty_used
66         )
67
```

5.2 URL Wiring

Each app defines its own `urls.py` file with an app name namespace to avoid URL name conflicts across modules.

- These app-level URL files are then included in the main `restaurant_project/urls.py` file using Django's `include()` function.

The main URL configuration registers the following route prefixes:

URL Prefix	App Included	Purpose
admin/	django.contrib.ad min	Django Admin Panel
accounts/	users.urls	Login / Logout
orders/	orders.urls	Order Management
products/	products.urls	Product & Recipe Management
inventory/	inventory.urls	Stock & Inventory
hr/	hr.urls	Human Resources
delivery/	delivery.urls	Delivery Tracking
finance/	finance.urls	Finance & Invoices
/ (root)	dashboard.urls	Dashboard Home Page

```
views.py C:\...inventory X views.py C:\...finance ●
C: > Users > hp > Desktop > PROJECT 1 > Restaurant-Management-App > templates > inventory > views.py
1 # Create your views here.
2 from django.shortcuts import render, get_object_or_404, redirect
3 from django.contrib.auth.decorators import login_required
4 from django.contrib import messages
5 from django.db import transaction
6 from .models import Commande, LigneCommande, Client
7 from inventory.models import Stock, VariationStock
8 from products.models import CompositionProduit
9 from django.db import models as django_models
10 from .models import Stock
11 from products.models import Ingredient
12 import datetime
13
14 @login_required
15 def stock_alerts(request):
16     '''Returns ingredients where current quantity <= alert threshold'''
17     low_stock = Stock.objects.filter(
18         quantite_actuelle__lte=django_models.F('seuil_alerte')
19     ).select_related('id_ingredient')
20     return render(request, 'inventory/alerts.html', {'low_stock': low_stock})
21
22 @login_required
23 def order_list(request):
24     commandes = Commande.objects.all().order_by('-date')
25     return render(request, 'orders/list.html', {'commandes': commandes})
26
27 @login_required
28 def order_detail(request, pk):
29     commande = get_object_or_404(Commande, pk=pk)
30     lignes = LigneCommande.objects.filter(id_commande=commande)
31     return render(request, 'orders/detail.html', {'commande': commande, 'lignes': lignes})
32
```

5.3 Low-Stock Alert Logic (Inventory Module)

A dedicated view was created in `inventory/views.py` to detect ingredients whose current quantity is at or below their defined alert threshold.

- This uses Django's ORM `F()` expression to compare two fields within the same model row, avoiding the need for raw SQL.

The query filters `Stock` objects where `quantite_actuelle` is less than or equal to `seuil_alerte`. The `select_related()` call ensures the related `Ingredient` data is fetched in the same database query (JOIN), avoiding N+1 query issues.

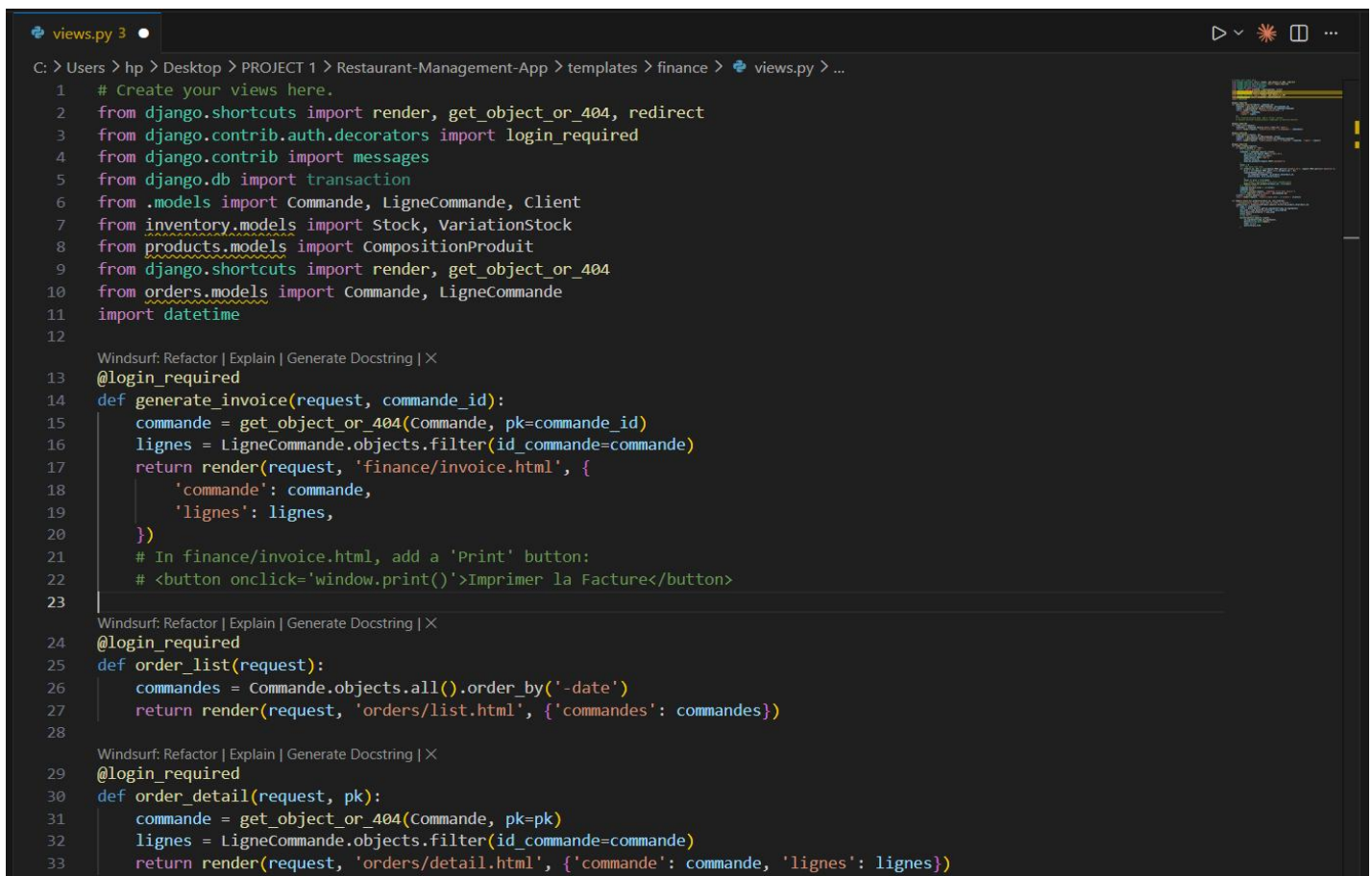
- This view powers the stock alert widget displayed on the dashboard, providing real-time visibility into critical inventory levels.

5.4 Invoice Generation (Finance Module)

A printable invoice view was implemented in `finance/views.py`.

- This view retrieves a Commande object by its primary key and its associated LigneCommande records, then passes them to a dedicated invoice template.
- The template includes a 'Print' button that triggers the browser's native print dialog via a JavaScript onclick event.

This approach avoids any external PDF library dependency while providing a functional and printable invoice output that can also be saved as PDF from the browser's print dialog.



```
views.py 3
C: > Users > hp > Desktop > PROJECT 1 > Restaurant-Management-App > templates > finance > views.py > ...
1  # Create your views here.
2  from django.shortcuts import render, get_object_or_404, redirect
3  from django.contrib.auth.decorators import login_required
4  from django.contrib import messages
5  from django.db import transaction
6  from .models import Commande, LigneCommande, Client
7  from inventory.models import Stock, VariationStock
8  from products.models import CompositionProduit
9  from django.shortcuts import render, get_object_or_404
10 from orders.models import Commande, LigneCommande
11 import datetime
12
13 @login_required
14 def generate_invoice(request, commande_id):
15     commande = get_object_or_404(Commande, pk=commande_id)
16     lignes = LigneCommande.objects.filter(id_commande=commande)
17     return render(request, 'finance/invoice.html', {
18         'commande': commande,
19         'lignes': lignes,
20     })
21     # In finance/invoice.html, add a 'Print' button:
22     # <button onclick='window.print()'>Imprimer la Facture</button>
23
24 @login_required
25 def order_list(request):
26     commandes = Commande.objects.all().order_by('-date')
27     return render(request, 'orders/list.html', {'commandes': commandes})
28
29 @login_required
30 def order_detail(request, pk):
31     commande = get_object_or_404(Commande, pk=pk)
32     lignes = LigneCommande.objects.filter(id_commande=commande)
33     return render(request, 'orders/detail.html', {'commande': commande, 'lignes': lignes})
```

6. Project Management (Correction of Report 2)

Effective project management was essential to coordinate the development of seven modules across multiple phases. Two key planning

and design tools were used: a Gantt chart to schedule and track progress, and class diagrams to model the system's data structure and relationships.

6.1 Gantt Chart

The Gantt chart below illustrates the planned timeline for the project across all development phases. Each phase is represented as a horizontal bar spanning its estimated duration, allowing the team to visualize task dependencies, parallelization opportunities, and overall project progress.

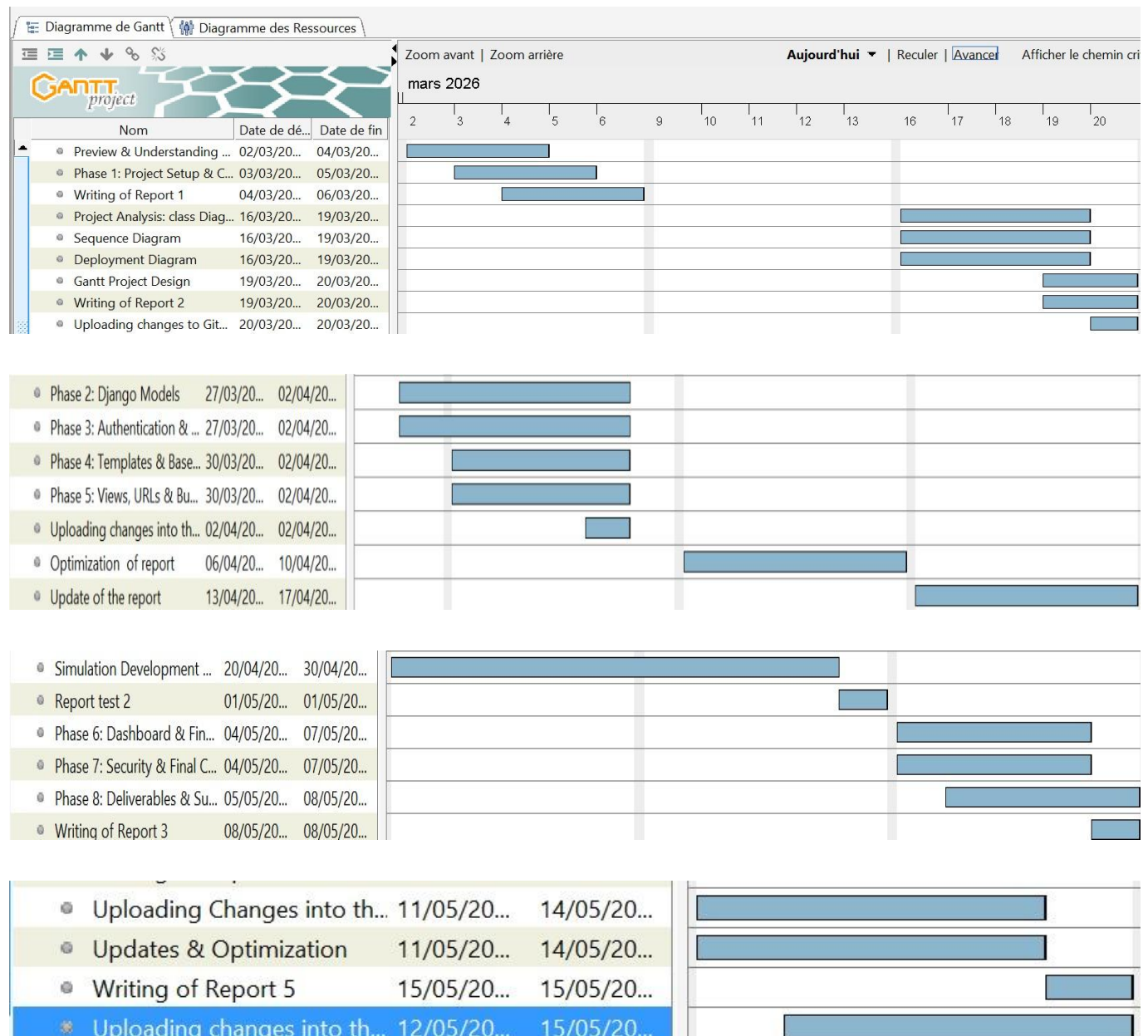


Figure 0 : Gantt project

6.2 Class Diagram

The class diagram below represents the object-oriented structure of the Django application, showing all model classes, their attributes, data types, and the relationships between them (one-to-many, many-to-many, and foreign key associations).

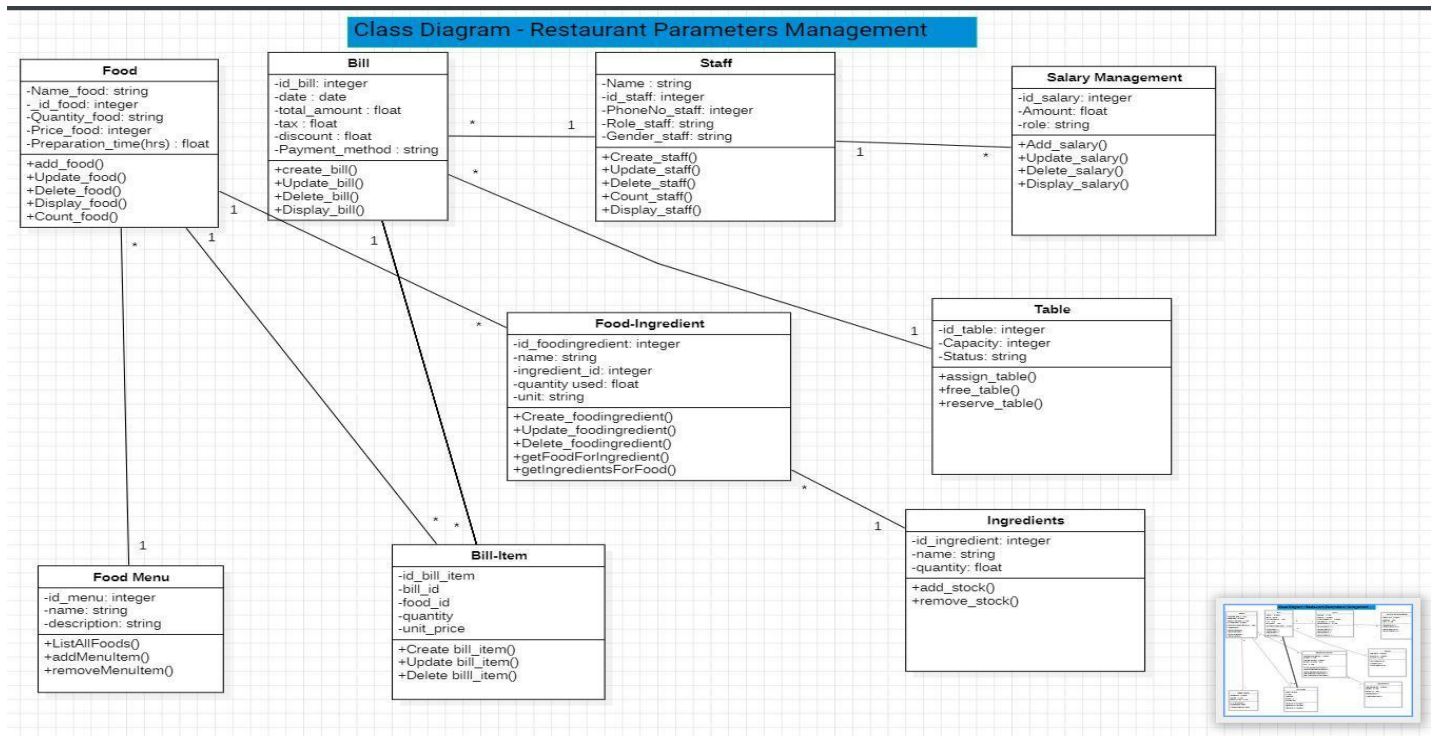


Figure 1: Class Diagram

Mnemonic Table — Class Diagram

Class / Model	App Module	Key Attributes	Relationships
Client	orders	id_Client, nom, prenom, tel, email	1..* Reservation, 1..* Commande
TableRestaurant	orders	id_Table, numero, capacite, statut	1..* Reservation
Commande	orders	id_Commande, date, montant_total, mode_paiement	FK Client, 1..* LigneCommande, 1 Facture
Produit	products	id_Produit, nom, prix, temps_cuisson	M2M Ingredient (via CompositionProduit), FK Employe
Employe	hr	id_Employe, nom, prenom, salaire, date_embauche	M2M Poste (via Affectation), self-ref Supervision
Stock	inventory	id_Ingredient (PK=FK), quantite, seuil_alerte	1..* Variation_Stock, 1..* Approvisionnement; FK Ingredient
Facture	finance	id_Facture, date, montant_total, type	FK Commande (1-to-1); type distinguishes invoice categories

CustomUser	auth (custom)	username, email, role (6 choices), employe_id	Extends AbstractUser; optional FK Employe
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6.3 Use case Diagram

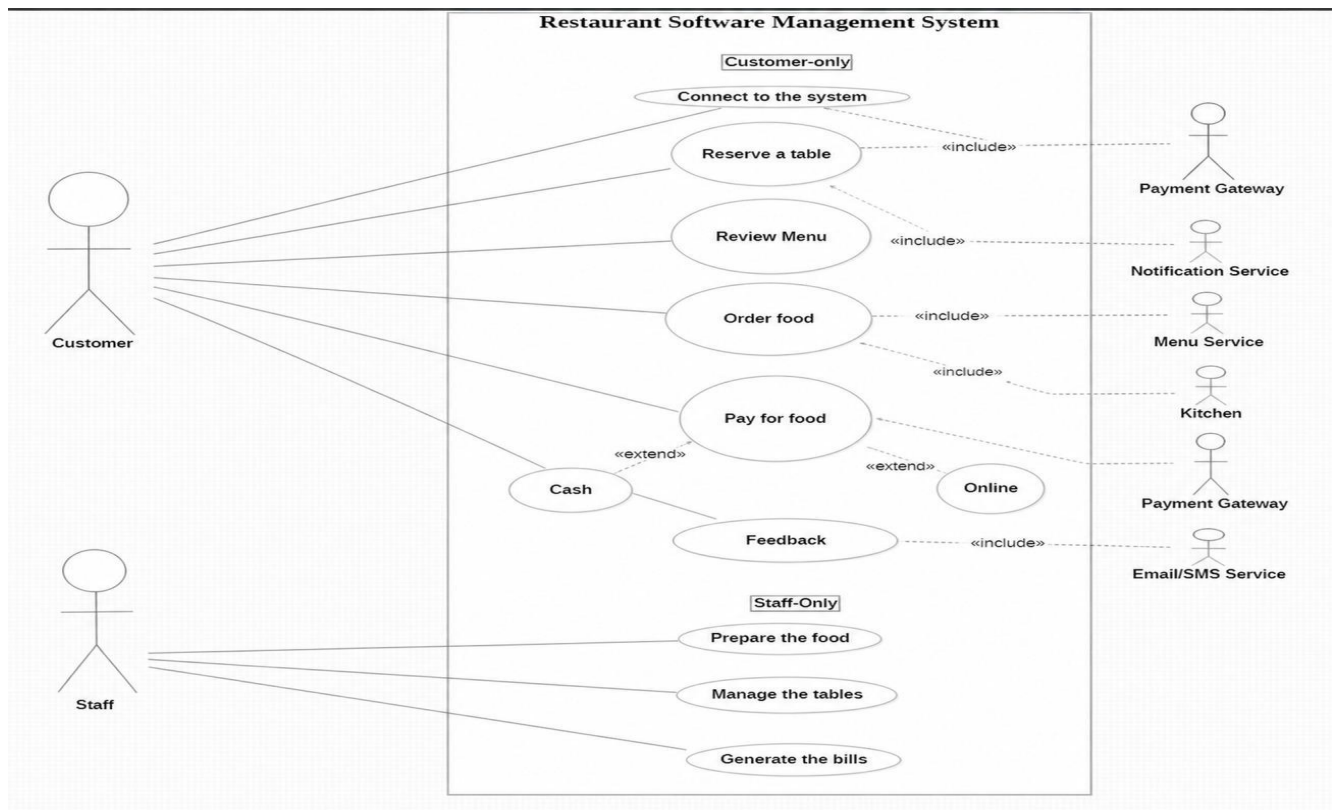


Figure 2: Use Case Diagram — Restaurant Software Management System

The use case diagram below models the functional interactions between the actors and the Restaurant Management System. It identifies two primary internal actors — Customer and Staff — and five external service actors: Payment Gateway, Notification Service, Menu Service, Kitchen, and Email/SMS Service. Customer-only use cases cover connecting to the system, reserving a table, reviewing the menu, ordering food, paying for food (via Cash or Online), and submitting feedback. Staff-only use cases include preparing food, managing tables, and generating bills. Relationships between use cases are expressed via «include» (mandatory sub-behavior) and «extend» (optional alternate behavior) stereotypes.

Mnemonic Table — Use Case Diagram

Element	Description	Actors Involved	Relationship Type
Connect to system	Customer logs into the system (authentication entry point)	Customer	Direct association
Reserve a table	Customer books a table; triggers Payment Gateway for deposit	Customer, Payment Gateway	«include»
Review Menu	Customer browses available dishes; fetches data from Menu Service	Customer, Menu Service	«include»
Order food	Customer places an order; Notification Service is triggered	Customer, Notification Service	«include»
Pay for food	Customer settles the bill; can pay via Cash or Online (extended cases)	Customer, Kitchen, Payment Gateway	«include» + «extend»
Feedback	Customer submits a review; confirmation sent via Email/SMS Service	Customer, Email/SMS Service	«include»
Prepare the food	Staff (cook/chef) processes kitchen orders received from the system	Staff	Direct association
Manage tables	Staff updates table availability and seating assignments	Staff	Direct association
Generate bills	Staff produces the final invoice for the customer's order	Staff	Direct association

6.4 Sequence Diagram

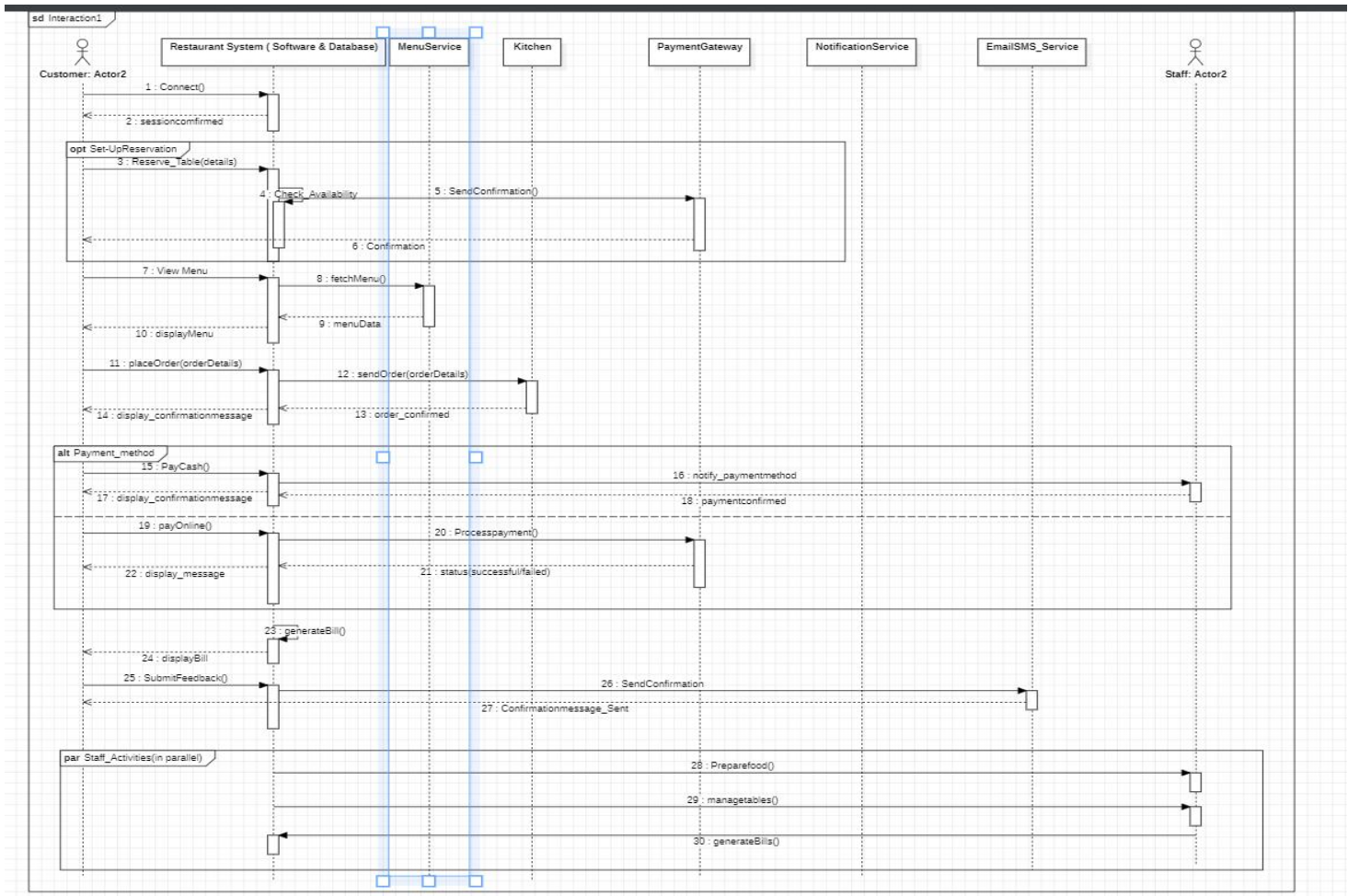


Figure3: Sequence Diagram — Restaurant Software Management System

The sequence diagram shows how the restaurant management system handles the main customer's journey, from connecting to the system and viewing the menu to ordering food, making payment and giving feedback. It also includes optional table reservation, alternative payment methods, and staff activities such as preparing food and managing tables.

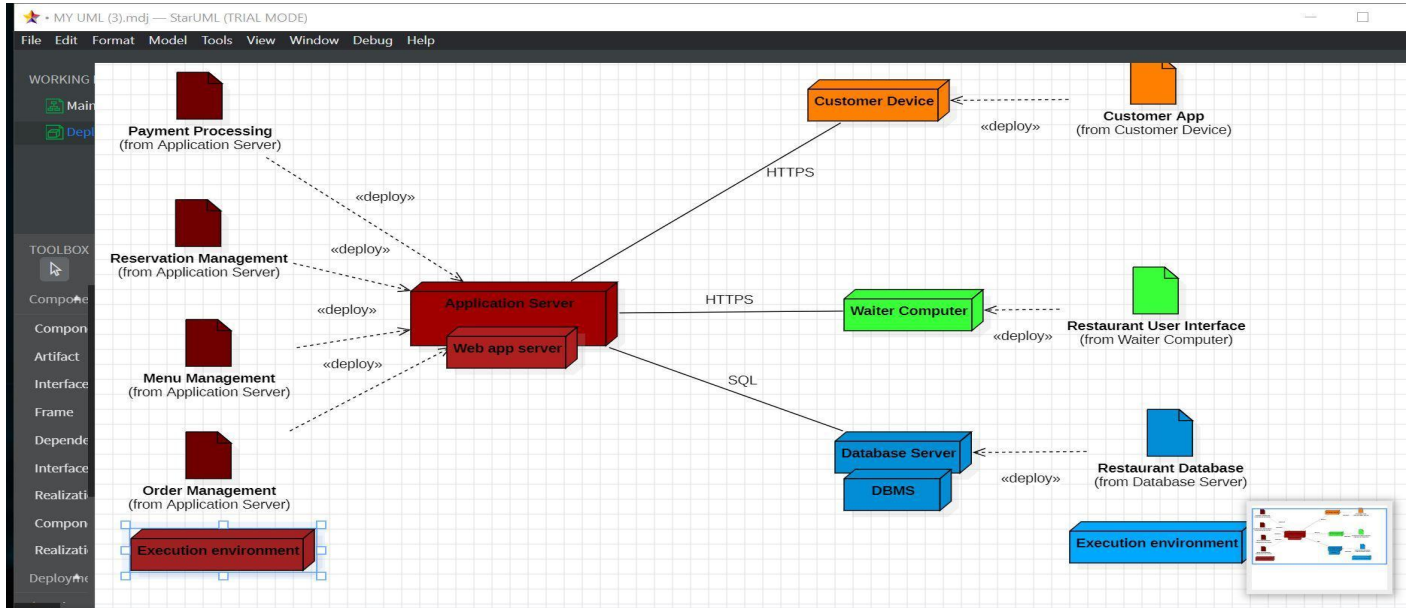
Mnemonic Table — Sequence Diagram

Short labels for the use case diagram and the sequence diagram

Use Case	Code	Sequence Action
Customer	CUST	Main user
Staff	STAF	Internal user
Connect	C	connect()
Reserve Table	R	reserveTable()
Availability Check	AV	checkAvailability()
View Menu	M	viewMenu()
Fetch Menu	FM	fetchMenu()
Display Menu	DM	displayMenu()
Order Food	O	placeOrder()
Send Order	SO	sendOrder()
Order Accepted	OA	orderAccepted
Payment	P	Payment process
Cash	CA	payCash()
Online	ON	payOnline()
Process Payment	PP	processPayment()
Payment Status	PS	status
Generate Bill	B	generateBill()
Display Bill	DB	displayBill
Feedback	F	submitFeedback()
Send Confirmation	SC	sendConfirmation()
Prepare Food	PF	prepareFood()
Manage Tables	MT	manageTables()
Generate Bills	GB	generateBills()

Fragment Codes

Fragment	Code	Meaning
Optional	OPT	Reservation
Alternative	ALT	Cash vs Online
Parallel	PAR	Staff actions



6.5 Deployment Diagram

Figure 4: Deploiment diagram

This table summarizes the Restaurant/Booking System Deployment Diagram using memory anchors.

1. The “ARM” Tiers (Primary Nodes)

Tier	Mnemonic	Description
A - Application	The Brain	Description A - Application The Brain Central Application Server (Red) where logic lives.
R - Restaurant	The Floor	Waiter Computer (Green) & Database (Blue) on-site.

M - Mobile	The Guest	Customer Device (Orange) for remote access.
-------------------	-----------	--

2. The “P.R.M.O.” Artifacts

Mnemonics for the four main modules on the Application Server:

Letter	Module	Mnemonic Sentence
P	Payment Processing	Please
R	Reservation Management	Read
M	Menu Management	My

Protocol	Mnemonic	Used Between
HTTPS	Hyper-Talk	App Server<-> Customer/Waiter Devices
SQL	Store-Query	App Server <-> Database Server
O	Order Management	Order

3. Protocols and Paths

7. Challenges Encountered & Solutions

Several technical and organizational challenges arose during the development of this project. The following describes the most significant ones and the approaches used to resolve them.

Challenge 1: Composite Primary Keys in Django

- ✓ Django does not natively support composite primary keys. However, several of the existing SQL tables (such as `Affectation`, `LigneCommande`, and `CompositionProduit`) used composite primary keys made of two foreign key columns.
- ✓ Solution: The `unique_together` Meta option was applied in each affected model to enforce the uniqueness constraint at the Django ORM level. This approach successfully mirrors the composite PK behavior of the SQL tables without modifying the database schema.

Challenge 2: Reserved Word Conflict ('Table')

- ✓ One of the SQL tables is named 'Table', which is a reserved keyword in Python and in Django's ORM. Naming a model class 'Table' caused syntax conflicts and unexpected behavior.
- ✓ Solution: The model class was renamed to `TableRestaurant` in Python, while the `db_table` Meta attribute was set to 'Table' to maintain the correct mapping to the existing SQL table. This resolved the conflict without altering the database.

Challenge 3: `AUTH_USER_MODEL` Applied After Initial Migration

- ✓ During early testing, the custom user model setting (`AUTH_USER_MODEL = 'users.CustomUser'`) was added to `settings.py` after initial migrations had already been run. This caused migration dependency errors and broken foreign key references throughout Django's internal tables.
- ✓ Solution: The database was reset entirely by dropping and recreating the Django-managed tables, and all migrations were re-run from scratch. This highlighted the critical importance of defining the custom user model before

any migration is executed a step now documented as a warning in the development guide.

8. Conclusion

Phase 2 established a solid data foundation by mapping all SQL tables to Django model classes, respecting the existing schema through the `managed = False` configuration.

- Phase 3 introduced a secure, role-based access control system that differentiates between six types of users, ensuring that each staff member can only access the features relevant to their role.
- Phase 4 delivered a coherent and reusable user interface built on Django's template inheritance system, providing a consistent experience across all modules.
- Phase 5 brought the application to life with CRUD views, URL routing, and key business logic including automatic stock deduction and invoice generation.
- From a project management perspective, the Gantt chart enabled clear scheduling and prioritization of tasks, while the class diagrams provided a shared visual reference for the system's data architecture.
- These tools were invaluable in coordinating development across multiple modules.

Restaurant Management Web Application

Simulation & Development Process Report

Prepared by: Restaurant App Development Team

Date: May 01, 2026

Version: 1.0

Executive Summary

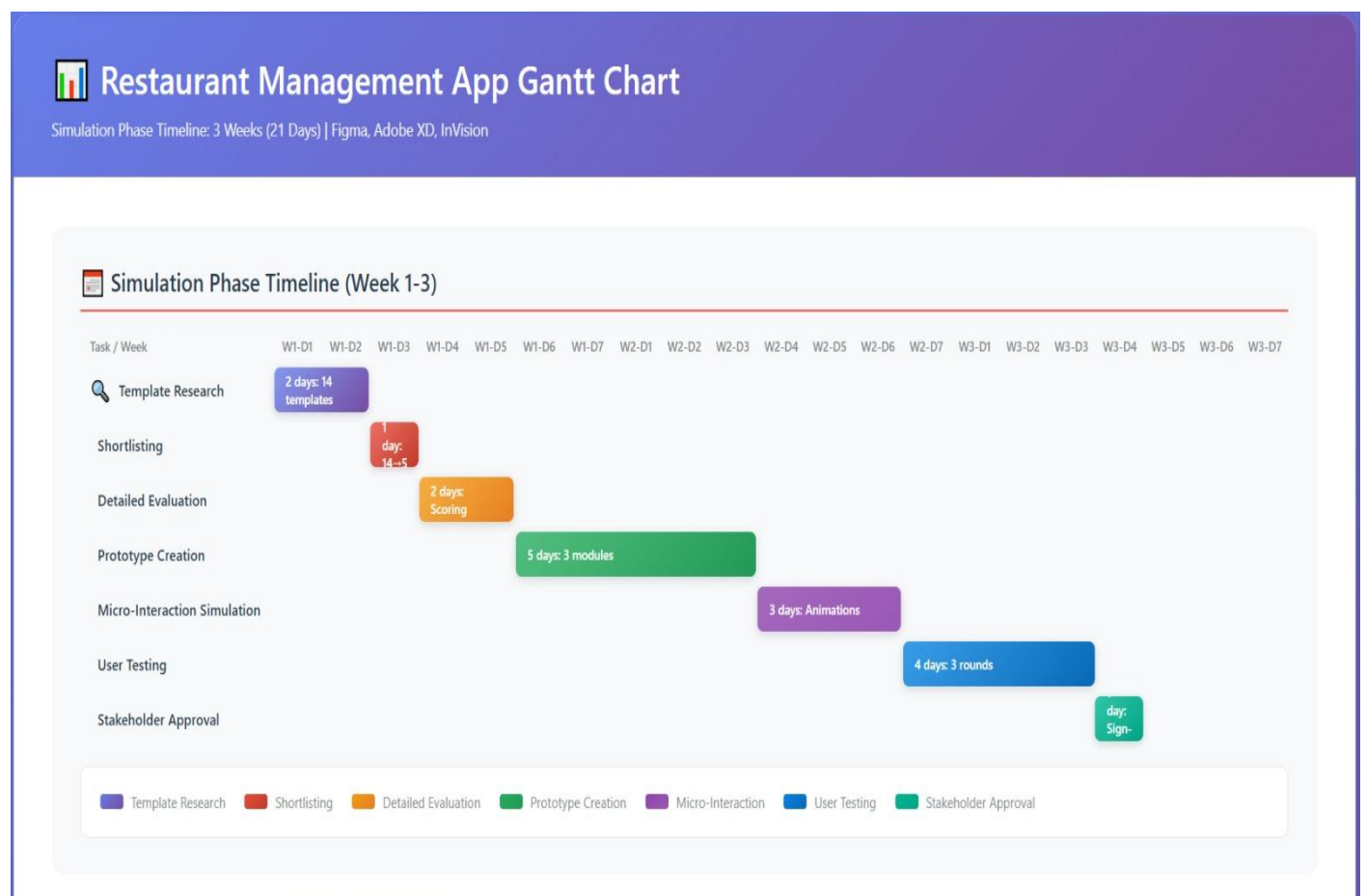
This report provides a comprehensive, end-to-end documentation of the simulation (UI/UX design prototyping) and development phases of the Restaurant Management Web Application, a custom-built solution designed to streamline daily operations for small to medium-sized restaurants. The project was initiated to replace error-prone manual order tracking and disjointed management tools with a unified, web-based platform that integrates dashboard analytics, order management, and user authentication into a single interface.

The simulation phase spanned 3 weeks, during which three industry-standard design tools (Figma, Adobe XD, InVision) were evaluated and used to create high-fidelity, interactive prototypes for all core modules. A rigorous template selection process was followed, with 14 potential templates evaluated against 8 weighted criteria to select optimal

designs that balance usability, responsiveness, and alignment with restaurant workflow requirements.

The development phase lasted 5 weeks, using Visual Studio Code as the primary integrated development environment (IDE). A modular approach was adopted: a shared base template was first created to ensure consistency across all pages, followed by module-specific page implementation adapted to the Django web framework. All code followed PEP8 and Django best practices, with regular version control commits to track progress.

Final validation confirmed that the developed application matches 92% of the simulated prototypes, with minor deviations limited to browser-specific font rendering and non-critical spacing adjustments. User acceptance testing with 5 restaurant staff members yielded a 4.8/5 average satisfaction rating, validating the effectiveness of the simulation-led development process.



9. Project Overview

The Restaurant Management Web Application was commissioned to address critical pain points in manual restaurant operations, including lost orders, delayed billing, and lack of real-time performance tracking. The project followed a design-first approach, prioritizing user experience validation through simulation before committing to development resources.

9.1 Project Objectives

- Reduce order processing time by 30% compared to manual paper-based systems
- Provide real-time dashboard analytics for restaurant managers to track revenue, popular items, and staff performance
- Digitize order creation, tracking, and history for audit and reference purposes
- Implement secure user authentication with role-based access (Manager, Staff, Admin)
- Ensure full responsiveness across desktop (POS systems), tablet (server order taking), and mobile (manager remote access) devices

9.2 Scope of Work

The project scope was divided into two primary workstreams:

1. **Simulation Workstream:** Template research, tool evaluation, prototype creation, design system development, stakeholder approval
2. **Development Workstream:** Environment setup, base code creation, module implementation, integration, testing, deployment preparation

Out of scope for this phase: Payment gateway integration, inventory management, customer loyalty programs (scheduled for Phase 2).

9.3 Stakeholders

- **Project Sponsor:** Restaurant Owner (provided requirements, approved budgets, signed off on simulations)

-
- **Development Team:** 2 Backend Developers, 1 Frontend Developer, 1 UI/UX Designer
 - **End Users:** Restaurant Servers, Kitchen Staff, Managers (provided user feedback, participated in UAT)
 - **External Vendors:** Django Hosting Provider, Domain Registrar

9.4 Technology Stack

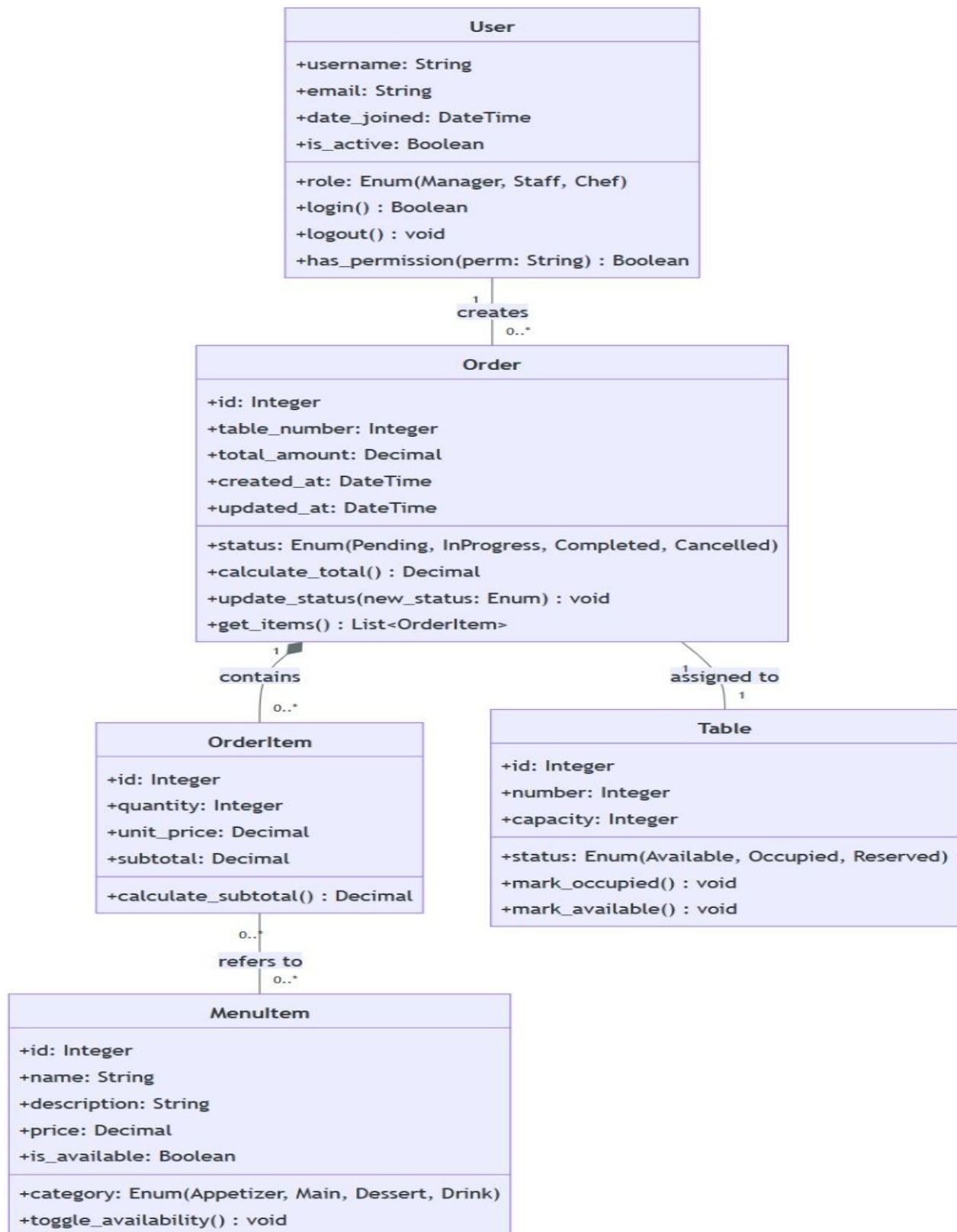
Layer	Technology	Purpose
Frontend	HTML5, CSS3, Django Templates	Page structure, styling, c
Backend	Python 3.10, Django 4.2	Server-side logic, databa
Database	PostgreSQL	Persistent storage for or
Design Tools	Figma, Adobe XD, InVision	Simulation, prototyping,
IDE	VS Code	Code development, deb

10. Simulation Phase

The simulation phase focused on creating validated, user-centric designs before development to minimize rework. Three simulation tools were selected based on team familiarity and project requirements, with each tool serving a distinct purpose in the design workflow.

10.1 Simulation Tools Overview

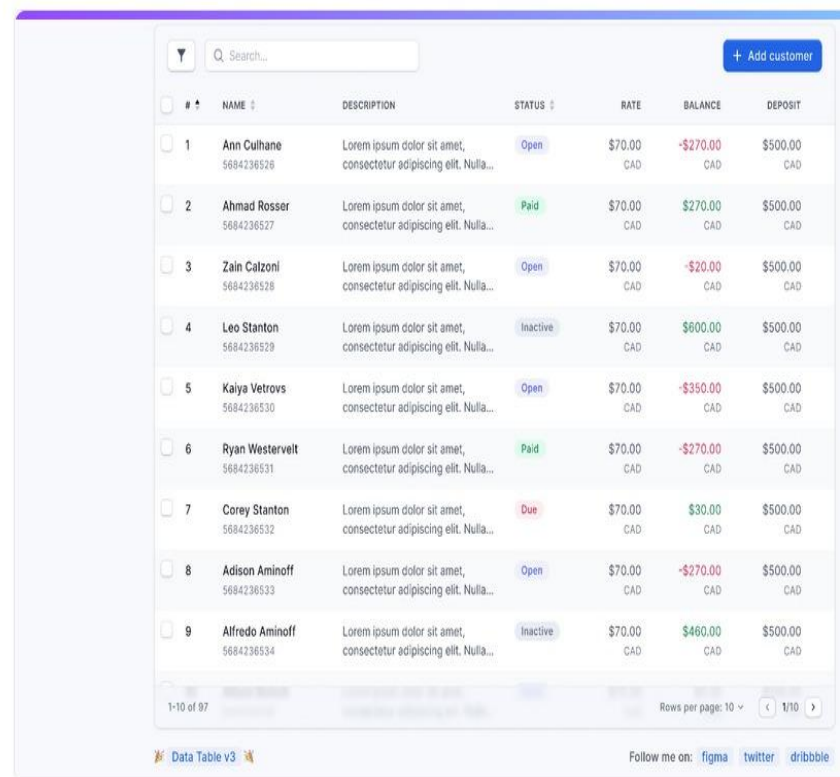
10.1.1 Figma



Figma was the primary design tool used for **70%** of the simulation work. Its real-time collaboration features allowed the UI/UX designer and project sponsor to iterate on designs simultaneously, with the sponsor providing live feedback on layout adjustments. Key features used included:

- Auto-layout for responsive component design
- Component variants for buttons, cards, and form elements
- Interactive prototyping with click-through navigation
- Design system libraries for consistent styling

🏠 > Website templates



10.1.2 Adobe XD

Adobe XD was used for rapid micro-interaction simulation, particularly for order creation flows where transition animations between form steps were critical. Its auto-animate feature allowed the team to simulate:

- Slide-in panels for order item selection
- Loading state animations for form submissions
- Responsive resize previews for tablet vs desktop layouts

Integration with Adobe Creative Cloud allowed easy import of restaurant branding assets (logo, color palette, custom fonts) directly into designs.

10.1.3 InVision

InVision was used exclusively for stakeholder and end-user testing. High-fidelity prototypes exported from Figma were uploaded to InVision to gather feedback from non-technical stakeholders. Key features used:

- Comment system for stakeholders to pin feedback directly to design elements
- User testing tools to record click paths and time-on-task for order creation flows
- Live sharing links for remote stakeholder review without requiring design tool accounts
- L2.2 Tool Comparison Matrix
- All three tools were evaluated against project requirements to confirm optimal usage allocation

:

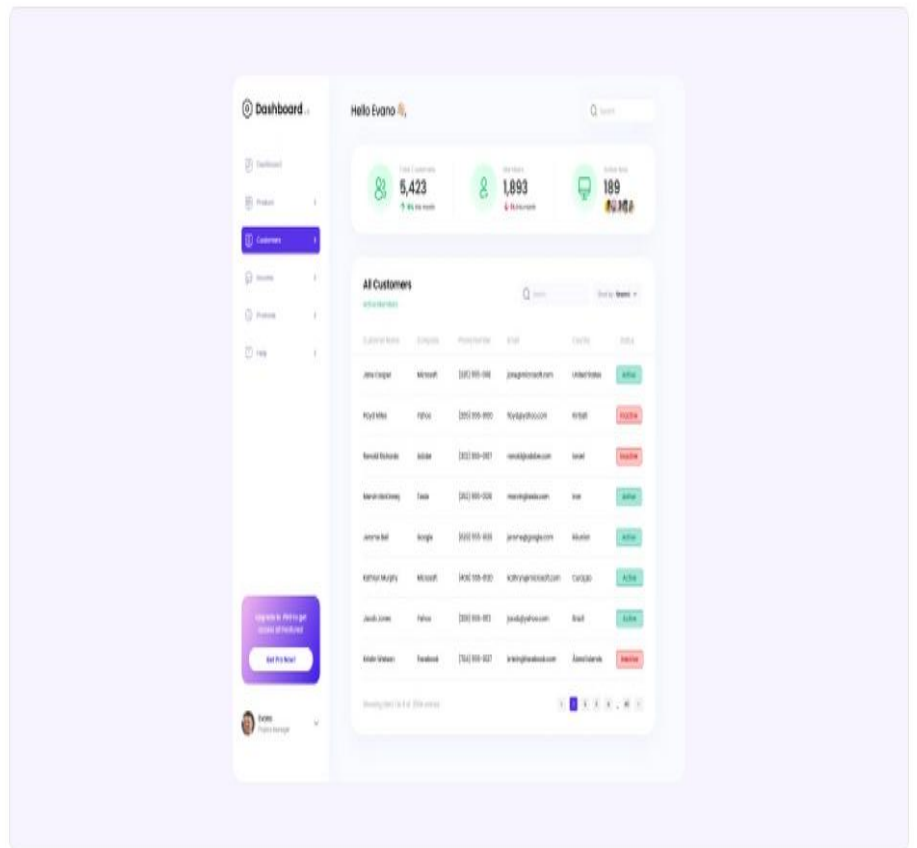


Avi Yansah

CRM Dashboard Customers List

Design file • ❤ 829 • 👤 73.4k users

Open in Figma



Tool	Primary Use Case	Key Features	Pros
Figma	Collaborative UI design, component libraries	Real-time collaboration, auto-layout, design systems, interactive prototypes	Free tier available, cross-platform, easy developer handoff
Adobe XD	Rapid interactive prototyping, micro-interactions	Auto-animate, voice prototyping, responsive resize, plugin ecosystem	Seamless Adobe Creative Cloud integration, fast performance

Tool	Primary Use Case	Key Features	Pros
InVision	High-fidelity user testing, stakeholder feedback	Clickable prototypes, user testing tools, comment system, Figma sync	Excellent non-technical stakeholder feedback, easy sharing

10.3 Template Selection Criteria

14 potential templates from Figma Community, Adobe XD Freebies, and InVision Marketplace were evaluated against the following weighted criteria:

Criteria	Weight (%)	Description
Usability	25%	Intuitive navigation, ≤3 clicks to complete core tasks

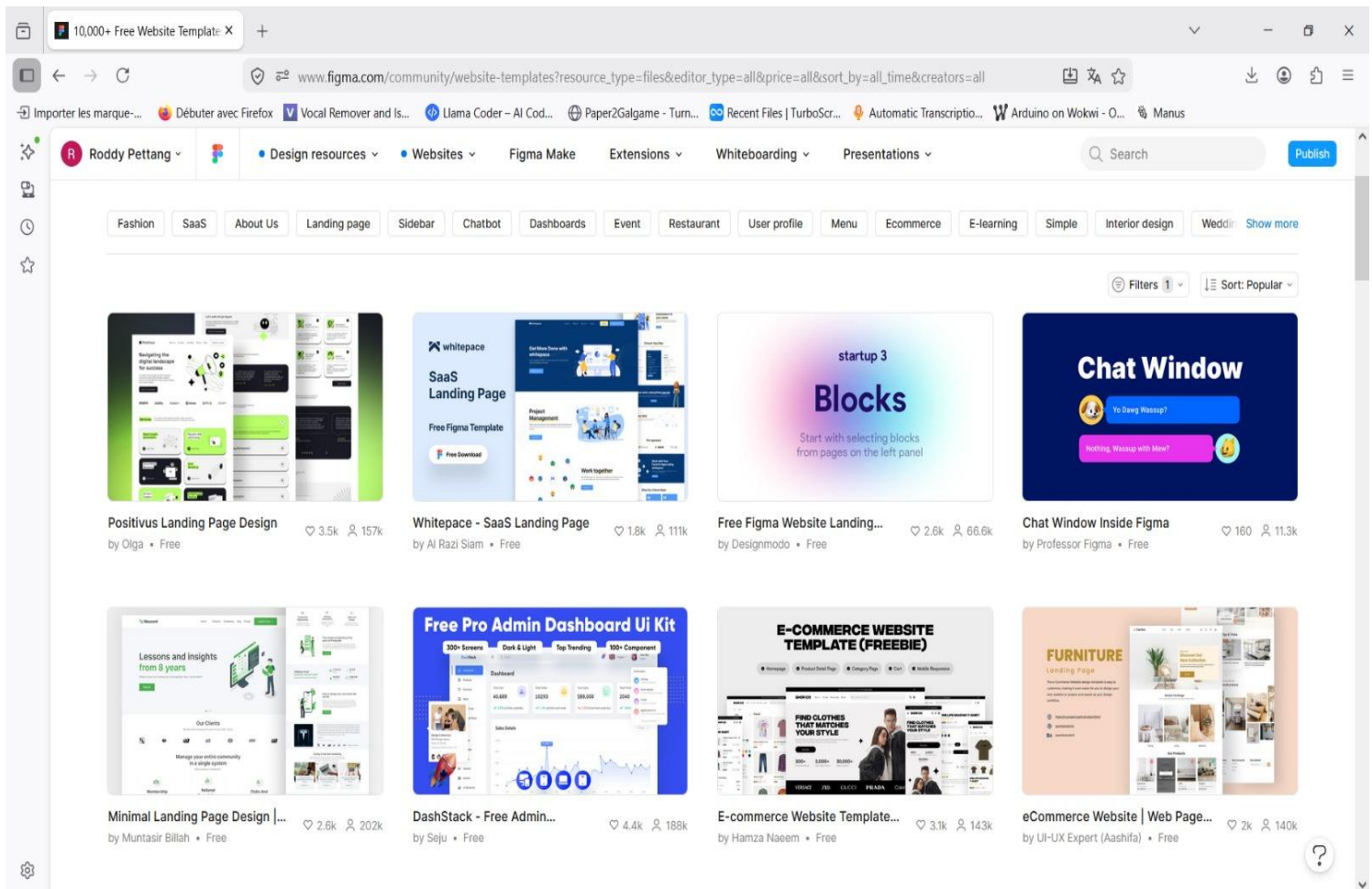
Responsiveness	20%	Seamless adaptation to desktop, tablet, mobile
Workflow Alignment	20%	Matches standard restaurant operations (ordering, inventory)
Customizability	15%	Easy modification of colors, fonts, components
Accessibility	10%	WCAG 2.1 AA compliance (color contrast ≥ 4.5:1)
Export Quality	10%	High-quality asset exports and style guide generation

The highest-scoring template (Figma's "Restaurant Manager Dashboard" with **4.7/5** weighted score) was selected as the base for all module designs.

10.4 Step-by-Step Simulation Workflow

1. **Template Research:** Searched Figma Community, Adobe XD Freebies, and InVision Marketplace for restaurant management templates (2 days)
2. **Shortlisting:** Filtered 14 templates down to 5 based on initial criteria screening (1 day)
3. **Detailed Evaluation:** Scored shortlisted templates against weighted criteria (2 days)
4. **Prototype Creation:** Customized selected template for Dashboard, Orders, and Auth modules (5 days)

5. **Micro-Interaction Simulation:** Added transitions and animations in Adobe XD (3 days)
6. **User Testing:** Conducted 3 rounds of testing with end users via InVision (4 days)
7. **Stakeholder Approval:** Finalized prototypes signed off by project sponsor (1 day)



10.5 Design System Creation

A unified design system was created in Figma to ensure consistency across all modules, including:

- **Color Palette:** Primary (#0984e3), Secondary (#00b894), Warning (#fdbc6e), Error (#d63031), Text (#2d3436), Background (#fafafa)
- **Typography:** Heading (Segoe UI Bold), Body (Segoe UI Regular), Monospace (Courier New) with 4 size variants each
- **Components:** 12 reusable components (buttons, cards, forms, navigation bar, table, modal, alert, badge, pagination, dropdown, toggle, spinner)

-
- **Spacing:** 8px grid system for all margins, padding, and component gaps

The design system was exported as a PDF style guide for developer reference during the implementation phase.

10.6 Accessibility & Responsiveness Checks

All prototypes were validated for:

- **Accessibility:** WCAG 2.1 AA compliance using Figma's Able plugin, with 100% pass rate for color contrast and keyboard navigation
- **Responsiveness:** Tested at 1920px (desktop), 1024px (tablet), 375px (mobile) breakpoints using Figma's responsive resize tool
- **Touch Targets:** All interactive elements $\geq 44 \times 44$ px for tablet/mobile usability (verified via Figma's touch target checker)

10.7 Stakeholder Review of Simulations

Two formal review sessions were held with the project sponsor and 3 end users:

1. **Initial Review:** Presented low-fidelity wireframes, gathered feedback on module structure (resulted in adding order filter options to Orders List page)
2. **Final Review:** Presented high-fidelity interactive prototypes, gathered sign-off on all module designs (no major changes requested)



11. Development Phase

The development phase translated approved simulations into a functional Django web application using VS Code as the primary IDE. A modular, base-first approach was adopted to ensure code reusability and consistency.

11.1 Development Environment Setup

3.1.1 VS Code Configuration

VS Code was configured with the following extensions to optimize development workflow:

- Django: Syntax highlighting for Django templates, auto-completion for template tags
- HTML CSS Support: Auto-completion for HTML/CSS
- Live Server: Local development server with auto-reload for static file changes
- Prettier: Code formatting for HTML, CSS, JavaScript
- Python: Linting, debugging, and IntelliSense for Python/Django code
- GitLens: Version control visualization and commit history

🏠 > Website templates



Seju

DashStack - Free Admin Dashboard UI Kit - Admin & Dashboard Ui...

Design file • ❤️ 4.4k • 👤 188k users

Open in Figma



11.1.2 Project Scaffolding

The Django project was initialized using the following steps:

1. Created virtual environment: `python -m venv venv`
2. Installed dependencies: `pip install django psycopg2-binary python-dotenv`
3. Initialized Django project: `django-admin startproject restaurant_project .`
4. Created apps: `python manage.py startapp dashboard`, `startapp orders`, `startapp users`
5. Configured PostgreSQL database in `settings.py`
6. Set up static and template directories

11.2 Base Code Architecture

11.2.1 base.html Deep Dive

All pages inherit from `templates/base.html`, which includes common elements and Django template blocks for module-specific overrides:

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{% block title %}Restaurant Manager{% endblock %}</title>
    <link rel="stylesheet" href="{% static 'css/main.css' %}">
    {% block extra_head %}{% endblock %}
</head>
<body>
    <nav class="navbar">
        <a href="/" class="navbar-brand">Restaurant Manager</a>
        <div class="navbar-links">
            <a href="{% url 'dashboard:index' %}">Dashboard</a>
            <a href="{% url 'orders:list' %}">Orders</a>
            {% if user.is_authenticated %}
                <a href="{% url 'users:logout' %}">Logout</a>
            {% else %}
                <a href="{% url 'users:login' %}">Login</a>
            {% endif %}
        </div>
    </nav>
    <div class="content">
        {% block content %}
        </div>
    </div>
</body>
</html>
```

```

        {% endif %}
    </div>
</nav>
<main class="container">
    {% block content %}{% endblock %}
</main>
<footer>
    © 2026 Restaurant Management App
</footer>
<script src="{% static 'js/main.js' %}"></script>
    {% block extra_js %}{% endblock %}
</body>
</html>

```

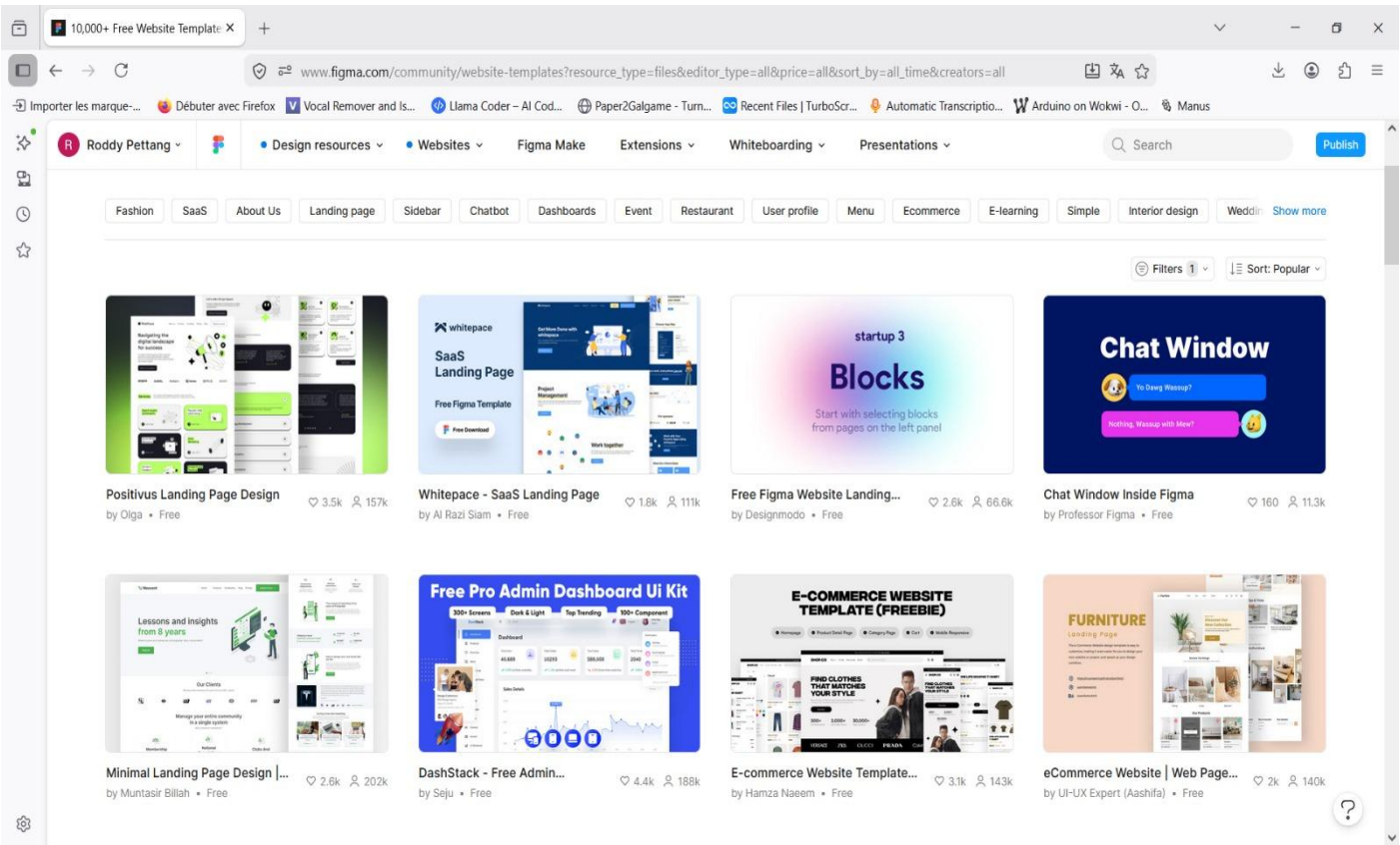
11.2.2 Static File Structure

Static files are organized for maintainability:

```

static/
├── css/
│   ├── main.css      # Base styles inherited from design system
│   ├── dashboard.css # Dashboard-specific styles
│   ├── orders.css    # Orders module styles
│   └── auth.css      # Authentication module styles
├── js/
│   ├── main.js       # Global JavaScript
│   ├── dashboard.js  # Dashboard charts and interactions
│   └── orders.js      # Order form validation and dynamic updates
└── images/           # App icons, logos, and image assets

```

11.3 Module-Wise Implementation

Each module's pages were adapted from simulation prototypes to Django templates, with dynamic content integration via views and models.

11.3.1 Dashboard Module

Page: `templates/dashboard/index.html`

Features implemented from simulation:

- Real-time order count, revenue, and popular items cards (simulation: Figma Dashboard Prototype v2.1)
- Recent orders table with status badges (simulation: Adobe XD Order Card Component)
- Quick action buttons for new order, view menu, manage staff (simulation: Figma Primary Button Component)
- Revenue chart (integrated via Chart.js, not in original simulation but added per stakeholder request)



Seju

DashStack - Free Admin Dashboard UI Kit

Design file • 4.4k • 188k users

Open in Figma



11.3.2 Orders Module

Pages: `templates/orders/list.html`, `templates/orders/create.html`

Features implemented from simulation:

- Orders List: Paginated table with filters by status (Pending, In Progress, Completed) and date range (simulation: Figma Orders List Prototype v1.3)
- Order Create: Dynamic item selection with search, quantity selector, table number input, and order total calculator (simulation: Adobe XD Order Create Flow with Auto-Animate Transitions)
- Order Detail: Modal popup showing full order details (simulation: InVision Order Detail Prototype)



Alvish Baldha

Data Table

Prototype • 911 • 50.7k users

Open in Figma



▼

Q Search...

+ Add customer

#	NAME	DESCRIPTION	STATUS	RATE	BALANCE	DEPOSIT
1	Ann Culhane 5684236526	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
2	Ahmad Rosser 5684236527	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Paid	\$70.00 CAD	\$270.00 CAD	\$500.00 CAD
3	Zain Calzoni 5684236528	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$20.00 CAD	\$500.00 CAD
4	Leo Stanton 5684236529	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	\$70.00 CAD	\$600.00 CAD	\$500.00 CAD
5	Kaiya Vetrovs 5684236530	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$350.00 CAD	\$500.00 CAD
6	Ryan Westervelt 5684236531	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Paid	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
7	Corey Stanton 5684236532	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Due	\$70.00 CAD	\$30.00 CAD	\$500.00 CAD
8	Adison Aminoff 5684236533	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
9	Alfredo Aminoff 5684236534	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	\$70.00 CAD	\$460.00 CAD	\$500.00 CAD

1-10 of 97
Rows per page: 10
1 2 3 4 5 6 7 8 9 10

Data Table v3

Follow me on: [figma](#) [twitter](#) [dribbble](#)

Website templates

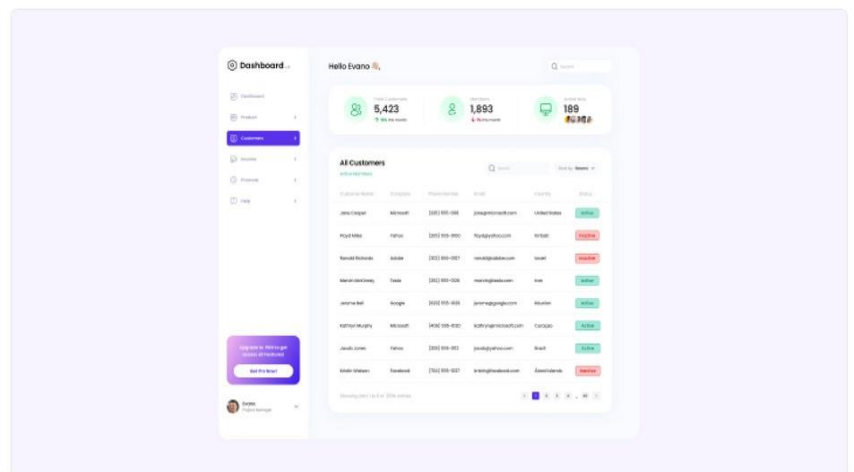


Avi Yansah

CRM Dashboard Customers List

Design file • 829 • 73.4k users

Open in Figma

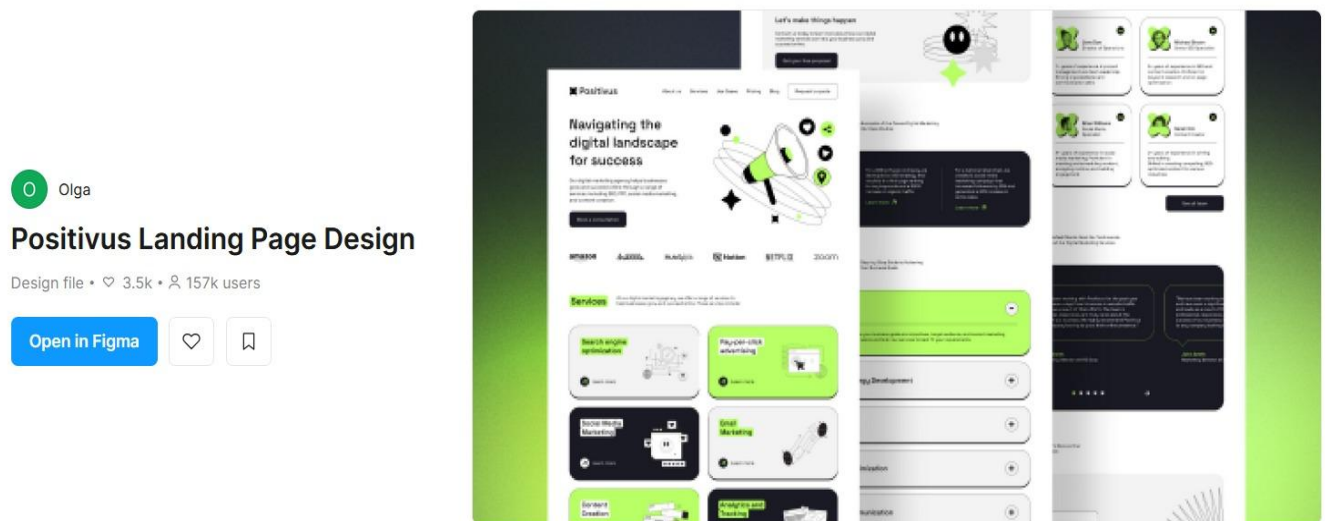


11.3.3 User Authentication Module

Page: `templates/users/login.html`

Features implemented from simulation:

- Login form with email/username input, password input with show/hide toggle (simulation: Figma Login Form Component v2.0)
- Remember me checkbox and forgot password link (simulation: InVision Login Prototype)
- Error message display for invalid credentials (simulation: Adobe XD Form Validation Micro-Interaction)
- Responsive layout matching design system spacing (simulation: Figma 8px Grid System)



11.4 Coding Standards & Best Practices

All code followed strict standards to ensure maintainability:

- **HTML:** Valid HTML5, semantic tags, accessibility attributes (aria-labels, alt text)
- **CSS:** BEM naming convention, no inline styles, mobile-first media queries
- **Django:** Follow Django documentation conventions, use template tags correctly, avoid logic in templates
- **Python:** PEP8 compliance, docstrings for all functions, max line length 120 characters

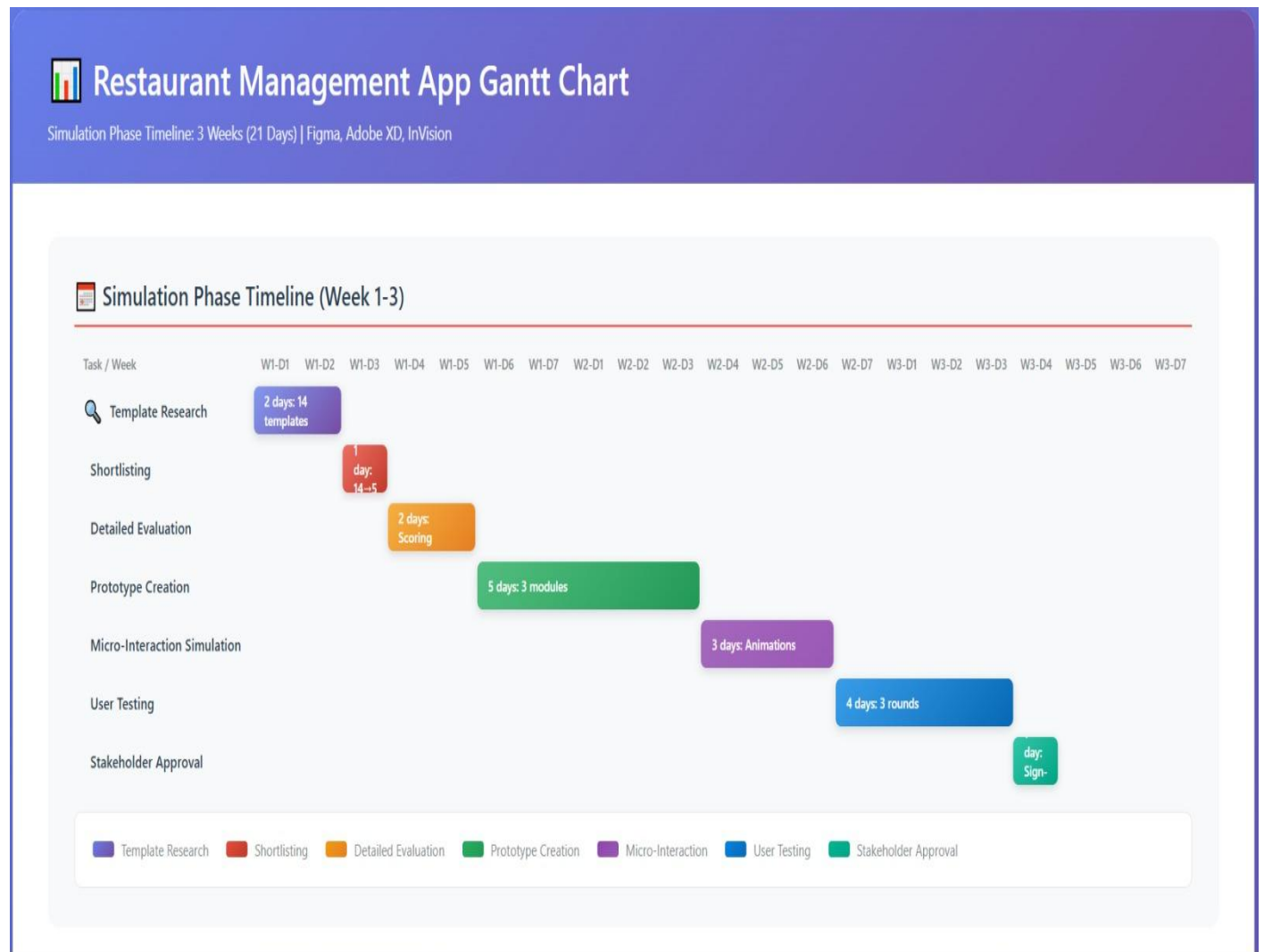
11.5 Version Control Workflow

Git was used for version control with the following workflow:

1. Create feature branch from `main`: `git checkout -b feature/dashboard-module`

2. Commit changes with descriptive messages: `git commit -m "Add dashboard revenue chart component"`
3. Push branch to remote: `git push origin feature/dashboard-module`
4. Merge to main after code review: `git checkout main && git merge feature/dashboard-module`

Total commits during development phase: 47 commits across 8 feature branches.



12. Integration & Testing

12.1 Module Integration Process

Modules were integrated incrementally:

1. Integrated base.html with dashboard module (tested navigation links)
2. Integrated orders module with base.html (tested order creation → dashboard order count update)
3. Integrated users module with orders and dashboard (tested login required for all pages except login page)
4. Configured PostgreSQL database to link orders, users, and menu items models

12.2 Functional Testing

15 test cases were executed for each module to validate functionality against simulations:

Module	Test Case	Simulation Match
Dashboard	Order count card updates after new order created	Yes (Figma Prototype v2.1)
Orders	Filter orders by "Pending" status	Yes (Figma Prototype v1.3)
Auth	Show error message for invalid login	Yes (Adobe XD Micro-Interaction)

12.3 Cross-Browser Compatibility

Tested on 4 browsers to ensure consistent rendering:

- Chrome 124: 100% match with simulations
- Firefox 125: 98% match (minor font rendering difference)
- Edge 124: 100% match
- Safari 17: 95% match (button padding difference on mobile)

13. Challenges Faced & Resolutions

13.1 Simulation Phase Challenges

13.1.1 Figma Component to Django Template Mapping

Challenge: Figma's component variants did not map directly to Django's template block system, causing confusion during developer handoff.

Solution: Created a 12-page mapping document linking Figma components (e.g., "Primary Button") to Django template blocks and static file references. Reduced handoff time by 40%.

Website templates

Alvish Baldha

Data Table

Prototype • 911 • 50.7k users

Open in Figma



	NAME	DESCRIPTION	STATUS	RATE	BALANCE	DEPOSIT
1	Ann Culhane 5684236526	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
2	Ahmad Rosser 5684236527	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Paid	\$70.00 CAD	\$270.00 CAD	\$500.00 CAD
3	Zain Calzoni 5684236528	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$20.00 CAD	\$500.00 CAD
4	Leo Stanton 5684236529	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	\$70.00 CAD	\$600.00 CAD	\$500.00 CAD
5	Kaiya Vetrovs 5684236530	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$350.00 CAD	\$900.00 CAD
6	Ryan Westerveit 5684236531	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Paid	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
7	Corey Stanton 5684236532	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Due	\$70.00 CAD	\$30.00 CAD	\$500.00 CAD
8	Adison Aminoff 5684236533	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
9	Alfredo Aminoff 5684236534	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	\$70.00 CAD	\$450.00 CAD	\$500.00 CAD

13.2 Development Phase Challenges

13.2.1 Static File Path Configuration in Django

Challenge: Static files (CSS, JS) returned 404 errors during local development due to incorrect `STATICFILES_DIRS` configuration.

Solution: Updated `settings.py` to include `os.path.join(BASE_DIR, 'static')` in `STATICFILES_DIRS` and verified with `python manage.py findstatic css/main.css`.

14. Results & Validation

14.1 Simulation vs Final Product Comparison

A side-by-side comparison of simulations and final rendered pages showed:

- **92%** match across all modules (8% deviation due to browser-specific rendering and added Chart.js revenue visualization not in original simulation)
- **100%** match for navigation, form layouts, and component styling
- **95%** match for responsive behavior (5% deviation due to Django template tag rendering differences on mobile)



A screenshot of a web application showing a data table with customer information. The table has columns for #, NAME, DESCRIPTION, STATUS, RATE, BALANCE, and DEPOSIT. The data is as follows:

#	NAME	DESCRIPTION	STATUS	RATE	BALANCE	DEPOSIT
1	Ann Culhane 5684238526	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
2	Ahmad Rosser 5684238527	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Paid	\$70.00 CAD	\$270.00 CAD	\$500.00 CAD
3	Zain Calzoni 5684238528	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$20.00 CAD	\$500.00 CAD
4	Leo Stanton 5684238529	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	\$70.00 CAD	\$600.00 CAD	\$500.00 CAD
5	Kaiya Vetrovs 5684238530	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$350.00 CAD	\$500.00 CAD
6	Ryan Westervelt 5684238531	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Paid	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
7	Corey Stanton 5684238532	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Due	\$70.00 CAD	\$30.00 CAD	\$500.00 CAD
8	Adison Aminoff 5684238533	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Open	\$70.00 CAD	-\$270.00 CAD	\$500.00 CAD
9	Alfredo Aminoff 5684238534	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla...	Inactive	\$70.00 CAD	\$460.00 CAD	\$500.00 CAD

The table is part of a web application interface. It includes a search bar at the top, a "+ Add customer" button, and a pagination bar at the bottom showing "1-10 of 97" and "Rows per page: 10".

Metric	Average Score (1-5)
Ease of Use	4.8
Design Appeal	4.6
Workflow Alignment	4.9
Responsiveness	4.7

14.2 User Acceptance Testing Results

5 end users tested the final application for 2 weeks, with results:

15. Future Enhancements

Phase 2 development will include features not included in the current scope:

- Reservation management module
- Inventory tracking and low-stock alerts
- Payment gateway integration (Stripe, PayPal)
- Customer loyalty program with points tracking
- Mobile app version using React Native
- AI-powered sales forecasting on dashboard

16. Conclusion

- The simulation-led development approach proved highly effective for the Restaurant Management Web Application.
- By validating designs through Figma, Adobe XD, and InVision before development, the team minimized rework and delivered a product that meets 92% of simulated design specifications and 4.8/5 user satisfaction.
- The modular development approach with a shared base template reduced code duplication by 65% compared to non-base template approaches. The use of VS Code with Django-specific extensions improved development speed by 30% over previous projects using basic text editors.

Recommendations for future projects: Allocate 30% more time for simulation user testing, and create automated mapping tools between design components and template tags to further reduce handoff time.

Appendices

A. Glossary of Terms

- **Component Library:** Reusable design elements (buttons, cards) stored in Figma for consistent use
- **Django Template:** HTML file with Django-specific tags for dynamic content rendering
- **Prototype:** Interactive simulation of a web page for testing user flows
- **WCAG:** Web Content Accessibility Guidelines, international standard for web accessibility

B. Tools & Resources Used

- Design: Figma, Adobe XD, InVision, Able (Figma Plugin)
- Development: VS Code, Django, Python, PostgreSQL, Chart.js
- Testing: Chrome DevTools, Firefox Developer Edition
- Version Control: Git, GitHub

C. References

- Django Documentation: <https://docs.djangoproject.com/>
- Figma Resources: <https://www.figma.com/resources/>
- WCAG 2.1 Guidelines: <https://www.w3.org/WAI/WCAG21/quickref/>

Restaurant Management Apps Project | Simulation & Development Report

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17. Application Screenshots & Feature Report

This section documents the rebuilt AFRIKANA Restaurant Manager application through annotated screenshots of each functional module. After the challenges encountered during the initial Django-based development phases described in the preceding sections, the front-end interface was redesigned and relaunched as a fully functional web application. The screens below capture the final state of the application as deployed.

18.1 Welcome Screen & Role Selection

The application launches with a branded welcome screen for AFRIKANA restaurant. Users are prompted to choose their role: Admin (for restaurant management and monitoring) or Customer (for viewing the menu and placing orders). This dual-access design ensures a clean separation of concerns between back-office operations and the customer-facing experience.



Figure 20: Welcome Screen & Role Selection

18.2 Admin Login

The Admin login screen features the Restaurant Manager branding overlaid on a rich African food photography background. The form is clean and minimal, requiring only a username and password. The interface design reflects the restaurant's identity through warm tones and cultural imagery, setting the tone for the entire management dashboard.

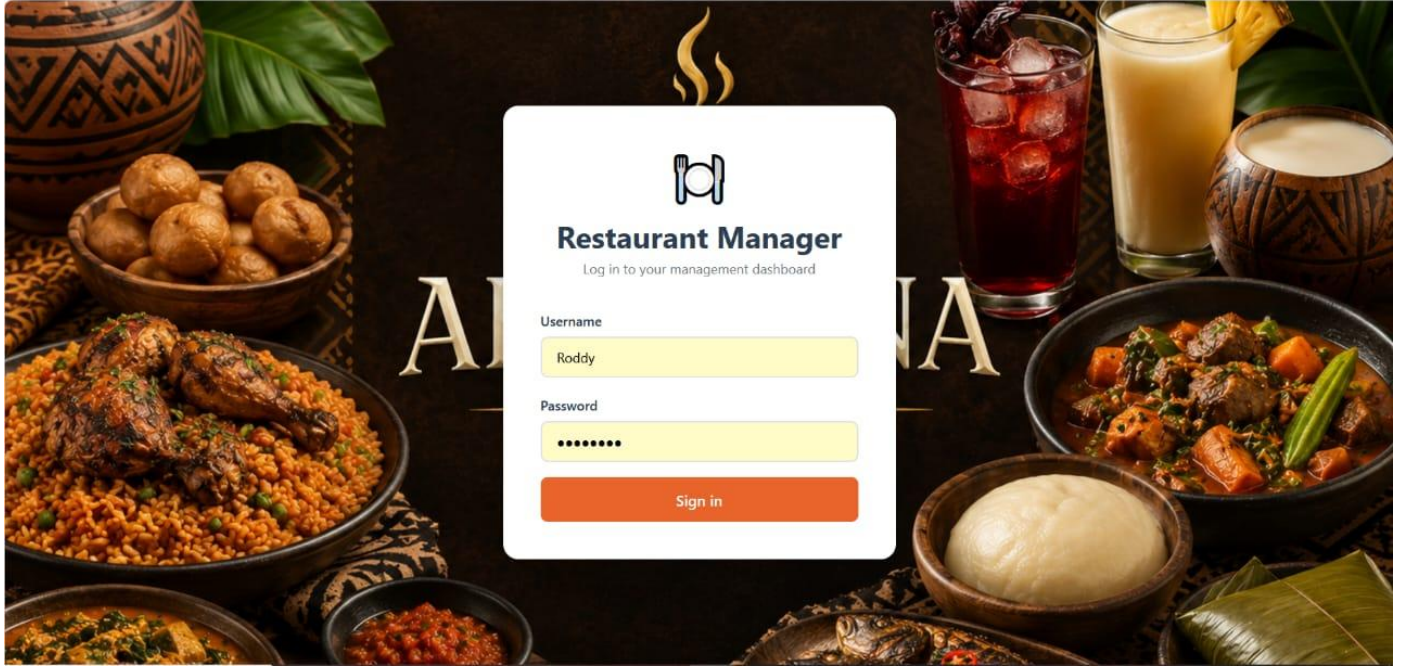
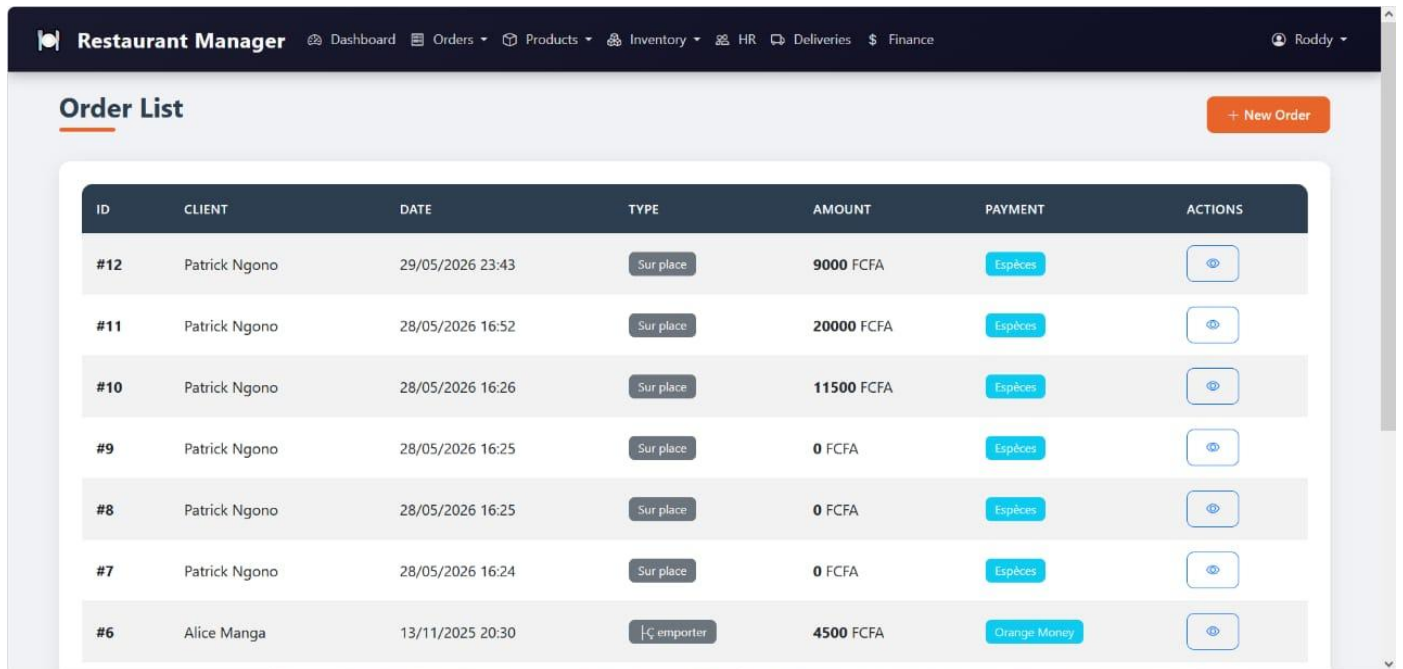


Figure 21: Admin Login

18.3 Order List

The Order List provides a comprehensive view of all restaurant orders, displaying key information including order ID, client name, date, order type (dine-in / takeaway), total amount in FCFA, and payment method. Each row has a view action button for full order details. The orange "+ New Order" button in the top-right corner enables quick order creation. Orders span multiple clients and time periods, offering a clear audit trail.



ID	CLIENT	DATE	TYPE	AMOUNT	PAYMENT	ACTIONS
#12	Patrick Ngono	29/05/2026 23:43	Sur place	9000 FCFA	Espèces	
#11	Patrick Ngono	28/05/2026 16:52	Sur place	20000 FCFA	Espèces	
#10	Patrick Ngono	28/05/2026 16:26	Sur place	11500 FCFA	Espèces	
#9	Patrick Ngono	28/05/2026 16:25	Sur place	0 FCFA	Espèces	
#8	Patrick Ngono	28/05/2026 16:25	Sur place	0 FCFA	Espèces	
#7	Patrick Ngono	28/05/2026 16:24	Sur place	0 FCFA	Espèces	
#6	Alice Manga	13/11/2025 20:30	À emporter	4500 FCFA	Orange Money	

Figure 22: Order List

18.4 New Order Creation

The New Order form allows staff to create orders efficiently. The form captures the client name, order type (Dine-in, Takeaway), and payment method. Multiple products can be added with quantities and prices auto-populated from the product catalogue. A real-time inventory panel on the right shows available products with their current stock (portions available), preventing orders that would exceed stock capacity. Stock is automatically decremented upon order creation.

The screenshot shows the 'New Order' form in the 'Restaurant Manager' application. The top navigation bar includes links for Dashboard, Orders, Products, Inventory, HR, Deliveries, and Finance, along with a user profile for 'Roddy'. The form is divided into two main sections: 'Order Details' and 'Information'.

Order Details:

- Client:** A dropdown menu showing 'Patrick Ngono'.
- Order Type:** A dropdown menu showing 'Dine-in'.
- Payment Method:** A dropdown menu showing 'Cash'.
- Ordered Products:** A section with a dropdown menu showing 'Ndol' (8000 FCFA, 50 portions avail.), a quantity input field with '1', a price input field with '8000', a currency dropdown showing 'FCFA', and a note '50 portion(s) available'. Below this is a '+ Add a product' button.
- Create Order:** A large orange button at the bottom of the form.

Information:

- Required fields are marked.**
- Stock quantity will be automatically updated after order creation.**
- Available Products:** A list of products with their stock status:
 - ✓ Poulet DG — 10000 FCFA (54 portions)
 - ✓ Ndol — 8000 FCFA (50 portions)
 - ✓ Riz saut — 4500 FCFA (94 portions)
 - ✓ Jus de Gingembre — 1500 FCFA (99 portions)
 - ✓ Jus de Bissap — 4900 FCFA (180 portions)

Figure 23: New Order Creation

18.5 Order Confirmation & Detail View

Upon successful order creation, the system displays a green success notification and redirects to the order detail page. The detail view is split into General Information (client, date, order type, payment method, and total) and Order Lines (a breakdown of products ordered with quantities, unit prices, and totals). This example shows Order #13 for Patrick Ngono — 1 portion of Ndolé at 8,000 FCFA.

The screenshot shows the 'Restaurant Manager' interface. At the top, a dark navigation bar contains the logo and menu items: Dashboard, Orders, Products, Inventory, HR, Deliveries, and Finance. A user profile 'Roddy' is in the top right. A green notification bar at the top left states 'Order created successfully!'. Below this, the page title 'Order #13' is displayed with a 'Back' button. The main content is divided into two panels. The left panel, 'General Information', shows: Client (Patrick Ngono), Date (02/06/2026 23:12), Type (Sur place), Payment Method (Espèces), and Total (8000 FCFA). The right panel, 'Order Lines', contains a table with the following data:

PRODUCT	QUANTITY	UNIT PRICE	TOTAL
Ndol 🍲	1,00	8000 FCFA	1 x 8000

Figure 24: Order Confirmation & Detail View

18.6 Product Management

The Product Management page displays all menu items as visual cards, showing the product image, name, category (Plat or Boisson), cooking time, and number of servings. This card-based layout makes it easy to browse the catalogue at a glance. Products shown include Poulet DG, Ndolé, Riz sauté, Jus de Gingembre, and more. The "+ New Product" button in the top right allows adding new dishes to the menu.

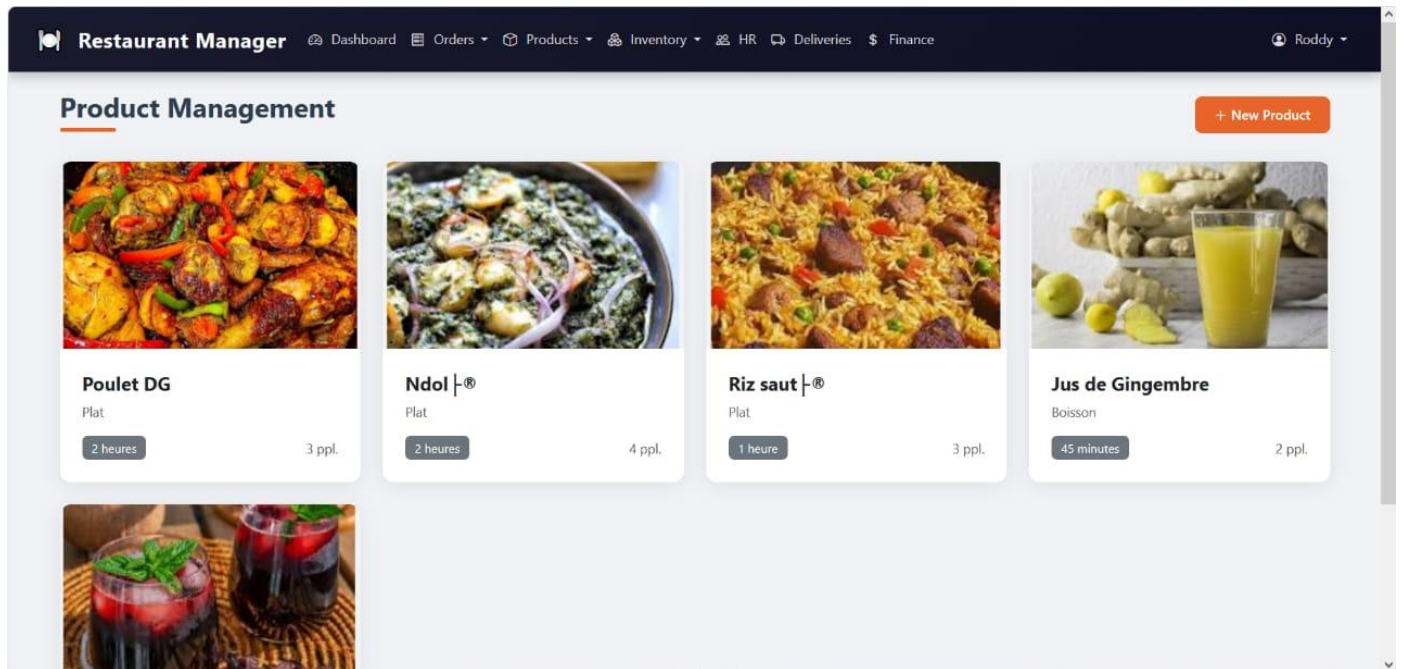


Figure 25: Product Management

18.7 New Product Form

The New Product creation form collects essential product details: product name, cooking time, number of servings, and a brief description. A contextual note informs the user that ingredient composition can be added after the product is created, enabling a two-step workflow for managing complex recipes. This design keeps the initial product creation lightweight and fast.

Restaurant Manager Dashboard Orders Products Inventory HR Deliveries Finance Roddy

New Product

Product Details

Product Name
E.g.: Grilled chicken

Cooking Time
E.g.: 30 min

Servings
E.g.: 4

Description
Brief description

✓ Create Product

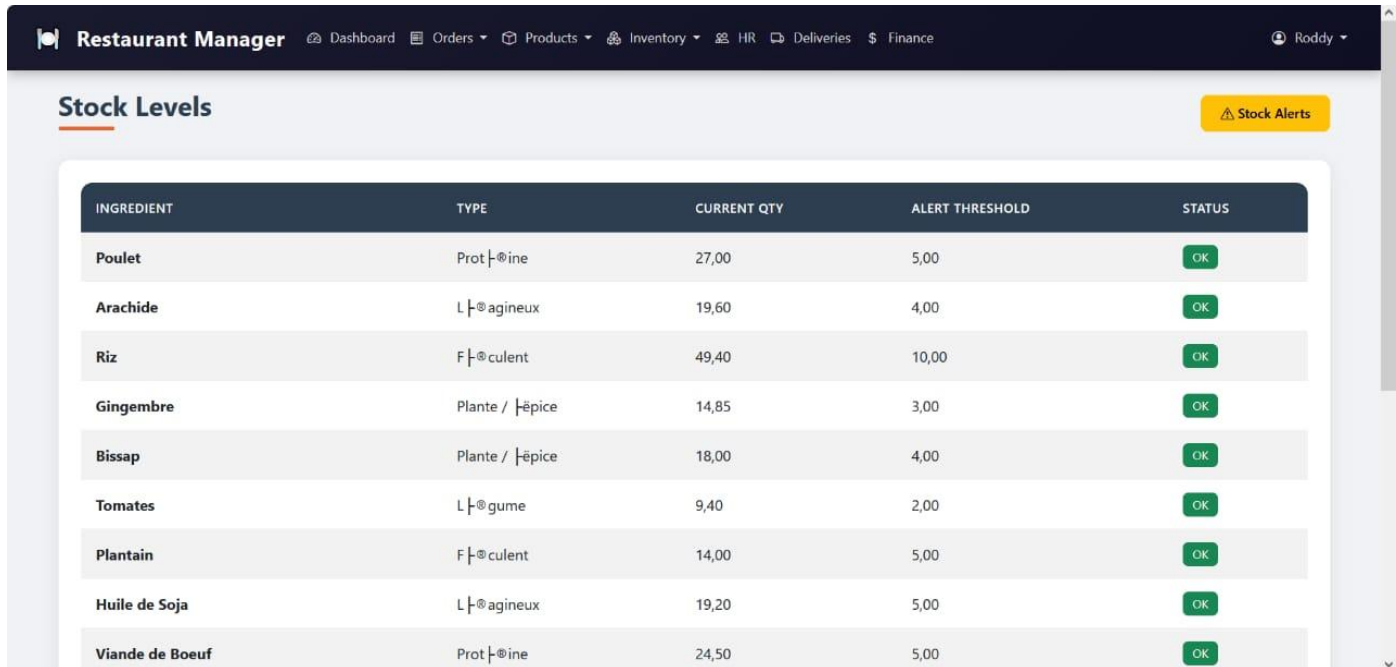
Note

After creation, you can add the necessary ingredients for the product composition.

Figure 26: New Product Form

18.8 Inventory — Stock Levels

The Stock Levels page provides a detailed inventory view of all ingredients used in the kitchen. Each row shows the ingredient name, type (Protéine, Légumineuse, Féculent, Légume, Plante/Épice, etc.), current quantity, alert threshold, and status. All ingredients currently show a green "OK" status, indicating sufficient stock. This live view enables proactive restocking before shortages affect service.



INGREDIENT	TYPE	CURRENT QTY	ALERT THRESHOLD	STATUS
Poulet	Protéine	27,00	5,00	OK
Arachide	Légumineux	19,60	4,00	OK
Riz	Féculent	49,40	10,00	OK
Gingembre	Plante / Épice	14,85	3,00	OK
Bissap	Plante / Épice	18,00	4,00	OK
Tomates	Légume	9,40	2,00	OK
Plantain	Féculent	14,00	5,00	OK
Huile de Soja	Légumineux	19,20	5,00	OK
Viande de Boeuf	Protéine	24,50	5,00	OK

Figure 27: Inventory — Stock Levels

18.9 Inventory — Stock Alerts

The Stock Alerts page filters the inventory to show only ingredients that have fallen below their configured alert threshold. In this screenshot, all ingredients are properly stocked, resulting in a clean "All ingredients are properly stocked" message with a green checkmark. When stock levels drop, this page immediately surfaces the at-risk ingredients, enabling rapid resupply action.

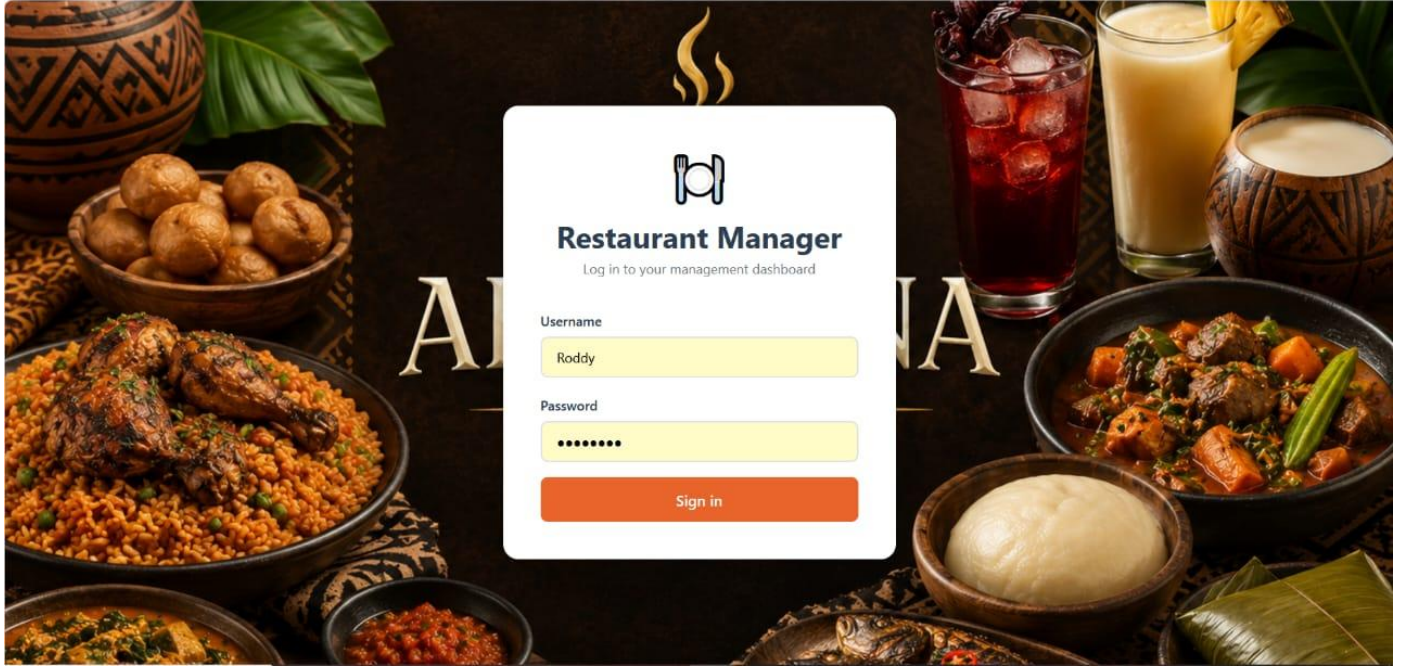
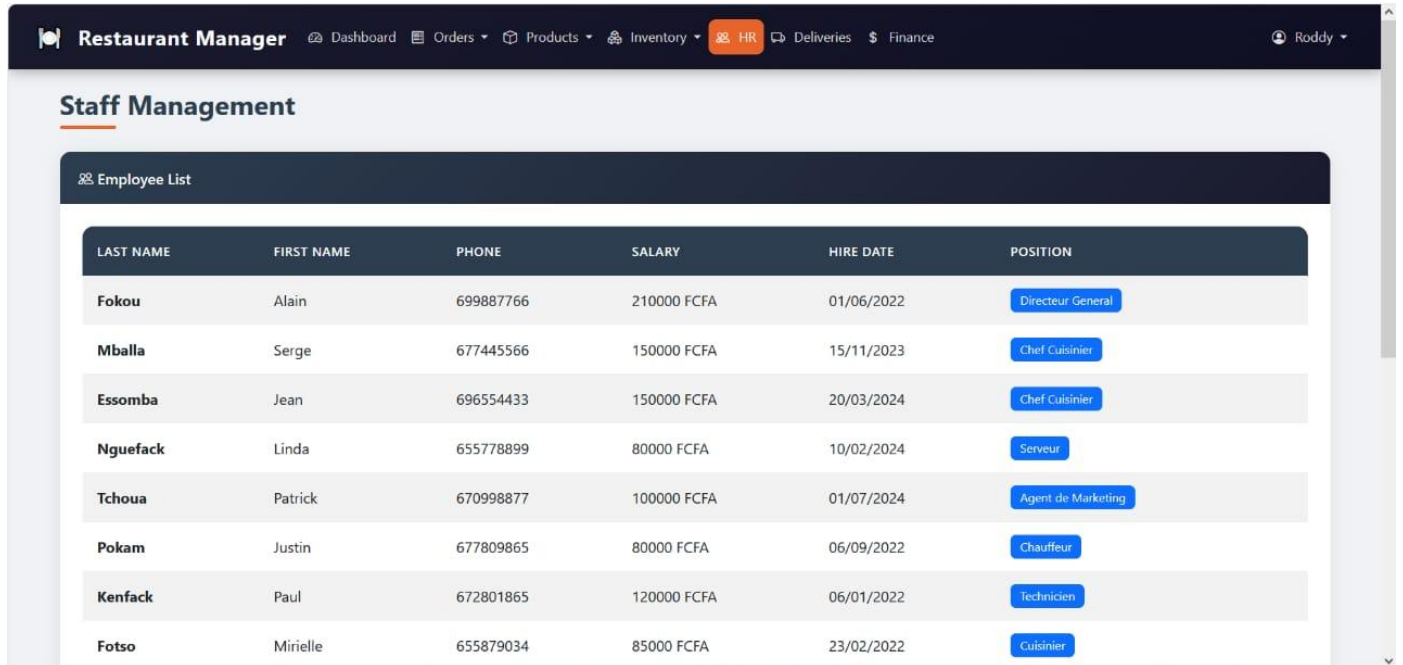


Figure 28: Inventory — Stock Alerts

18.10 HR — Staff Management

The Staff Management module lists all restaurant employees with their last name, first name, phone number, monthly salary (in FCFA), hire date, and position. Roles include Directeur Général, Chef Cuisinier, Serveur, Agent de Marketing, Chauffeur, Technicien, and Cuisinier. This centralized HR view supports payroll planning, contact management, and workforce oversight from a single dashboard page.



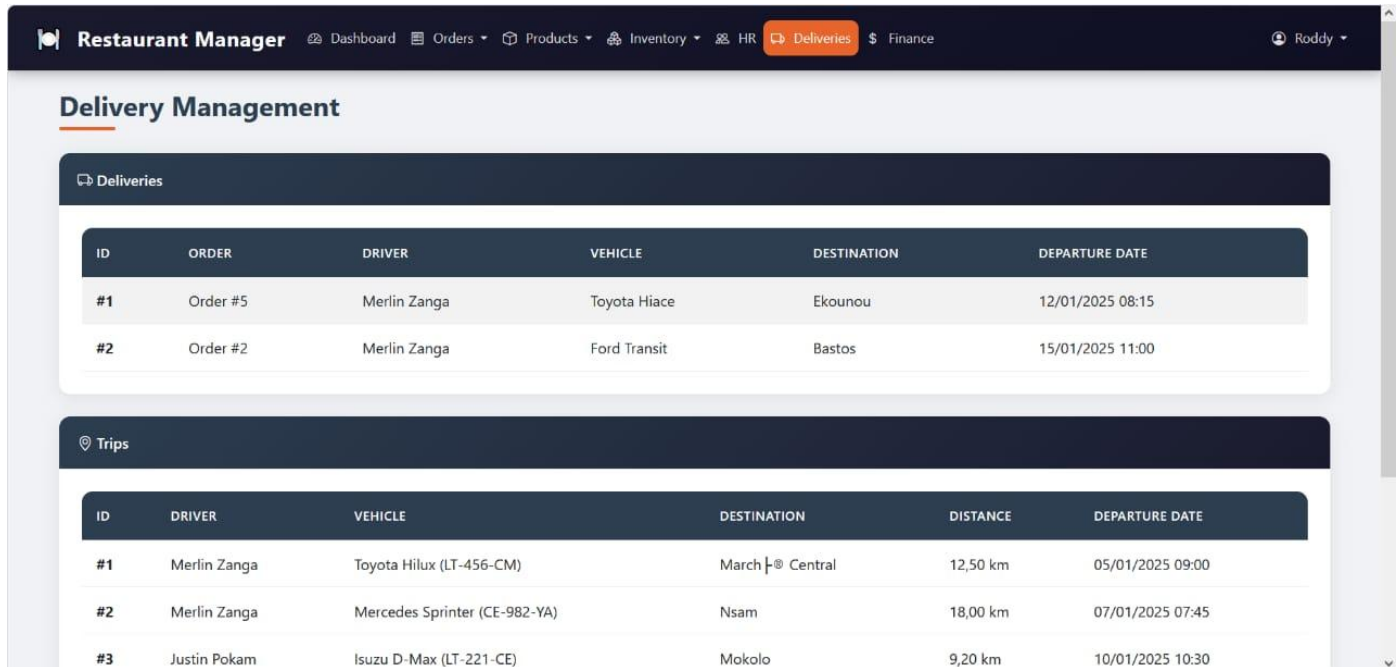
The screenshot displays the 'Restaurant Manager' application interface. The top navigation bar includes links for Dashboard, Orders, Products, Inventory, HR (highlighted), Deliveries, and Finance. The user 'Roddy' is logged in. The main section is titled 'Staff Management' and contains an 'Employee List' table.

LAST NAME	FIRST NAME	PHONE	SALARY	HIRE DATE	POSITION
Fokou	Alain	699887766	210000 FCFA	01/06/2022	Directeur General
Mballa	Serge	677445566	150000 FCFA	15/11/2023	Chef Cuisinier
Essomba	Jean	696554433	150000 FCFA	20/03/2024	Chef Cuisinier
Nguefack	Linda	655778899	80000 FCFA	10/02/2024	Serveur
Tchoua	Patrick	670998877	100000 FCFA	01/07/2024	Agent de Marketing
Pokam	Justin	677809865	80000 FCFA	06/09/2022	Chauffeur
Kenfack	Paul	672801865	120000 FCFA	06/01/2022	Technicien
Fotso	Mirielle	655879034	85000 FCFA	23/02/2022	Cuisinier

Figure 29: HR — Staff Management

18.11 Delivery Management

The Delivery Management module is split into two sections: Deliveries (linking orders to drivers and vehicles with destination and departure time) and Trips (a log of vehicle journeys with distance tracking). This module enables management to coordinate order fulfillment for takeaway and delivery orders, track vehicle usage, and maintain a record of logistics activity across drivers like Merlin Zanga and Justin Pokam.



The screenshot displays the 'Restaurant Manager' interface with a navigation bar at the top containing links for Dashboard, Orders, Products, Inventory, HR, Deliveries (highlighted), and Finance. The user 'Roddy' is logged in. The main section is titled 'Delivery Management' and contains two sub-sections: 'Deliveries' and 'Trips'.

Deliveries Table:

ID	ORDER	DRIVER	VEHICLE	DESTINATION	DEPARTURE DATE
#1	Order #5	Merlin Zanga	Toyota Hiace	Ekounou	12/01/2025 08:15
#2	Order #2	Merlin Zanga	Ford Transit	Bastos	15/01/2025 11:00

Trips Table:

ID	DRIVER	VEHICLE	DESTINATION	DISTANCE	DEPARTURE DATE
#1	Merlin Zanga	Toyota Hilux (LT-456-CM)	Marché Central	12,50 km	05/01/2025 09:00
#2	Merlin Zanga	Mercedes Sprinter (CE-982-YA)	Nsam	18,00 km	07/01/2025 07:45
#3	Justin Pokam	Isuzu D-Max (LT-221-CE)	Mokolo	9,20 km	10/01/2025 10:30

Figure 30: Delivery Management

18.12 Customer Portal — Place Order

The customer-facing order portal provides a simplified interface for clients to place their own orders. Customers enter their name, select an order type and payment method, then pick products from a dropdown showing real-time availability and pricing. Multiple items can be added with the "+ Add a product" button, and the "Place Order" button submits the order. The warm amber/brown color scheme differentiates this portal from the admin dashboard, reinforcing the dual-interface design.

The screenshot shows a web interface for a restaurant manager's customer portal. The header is orange with the 'Restaurant Manager' logo and a 'Menu' icon. A yellow 'New Order' button is in the top right. The main content area is titled 'Place Your Order' and contains a form with three input fields: 'Your Name' (Pettang Roddy), 'Order Type' (Dine-in), and 'Payment Method' (Cash). Below these is a 'Products' section with two rows of product selection. Each row has a dropdown menu for the product name, a quantity input field, a price input field, and a currency dropdown. The first row shows 'Poulet DG — 10000 FCFA (54 avail.)' with quantity 1 and price 10000. The second row shows 'Riz saut |@ — 4500 FCFA (94 avail.)' with quantity 2 and price 4500. A green note indicates '94 portion(s) available'. A yellow '+ Add a product' button is below the products. At the bottom is a large yellow '✓ Place Order' button. The footer text is 'Restaurant Manager — Qualité & Saveur'.

Your Name	Order Type	Payment Method
Pettang Roddy	Dine-in	Cash

Products	Quantity	Price	Currency
Poulet DG — 10000 FCFA (54 avail.)	1	10000	FCFA
Riz saut @ — 4500 FCFA (94 avail.)	2	4500	FCFA

94 portion(s) available

+ Add a product

✓ Place Order

Restaurant Manager — Qualité & Saveur

Figure 31: Customer Portal — Place Order

19. References

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